



UNIVERSITÀ DI VERONA

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USO DEGLI LLM PER ANALIZZARE E PREDIRRE TRAJETTORIE TURISTICHE

STUDENTE: Mattioli Simone

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INTRODUZIONE AL PROBLEMA

01



PREDIRRE



SFIDE ATTUALI

- Pattern Complessi
- Influenze Multiple

OPPORTUNITÀ

- LLM → Comprensione Contestuale
- Capacità Di Calcolo In Crescita

OBIETTIVI

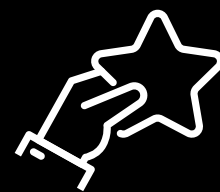
02

COSA SI VUOLE RAGGIUNGERE?



Reattività del sistema

Deve Essere Real-Time



FrameWork Indipendente dall' LLM

LLM Intercambiabile



Precisione Su Dati Reali

Test sulle VeronaCard

METODOLOGIA

03



RICERCA PRELIMINARE

FrameWork Pre-Esistenti



SVILUPPO

Nuove funzionalità



VALIDAZIONE

Feature Implementata



ANALISI PRESTAZIONI

Con Metriche Specifiche

PIANO STRATEGICO

01



STEP PRATICI EFFETTUATI

02



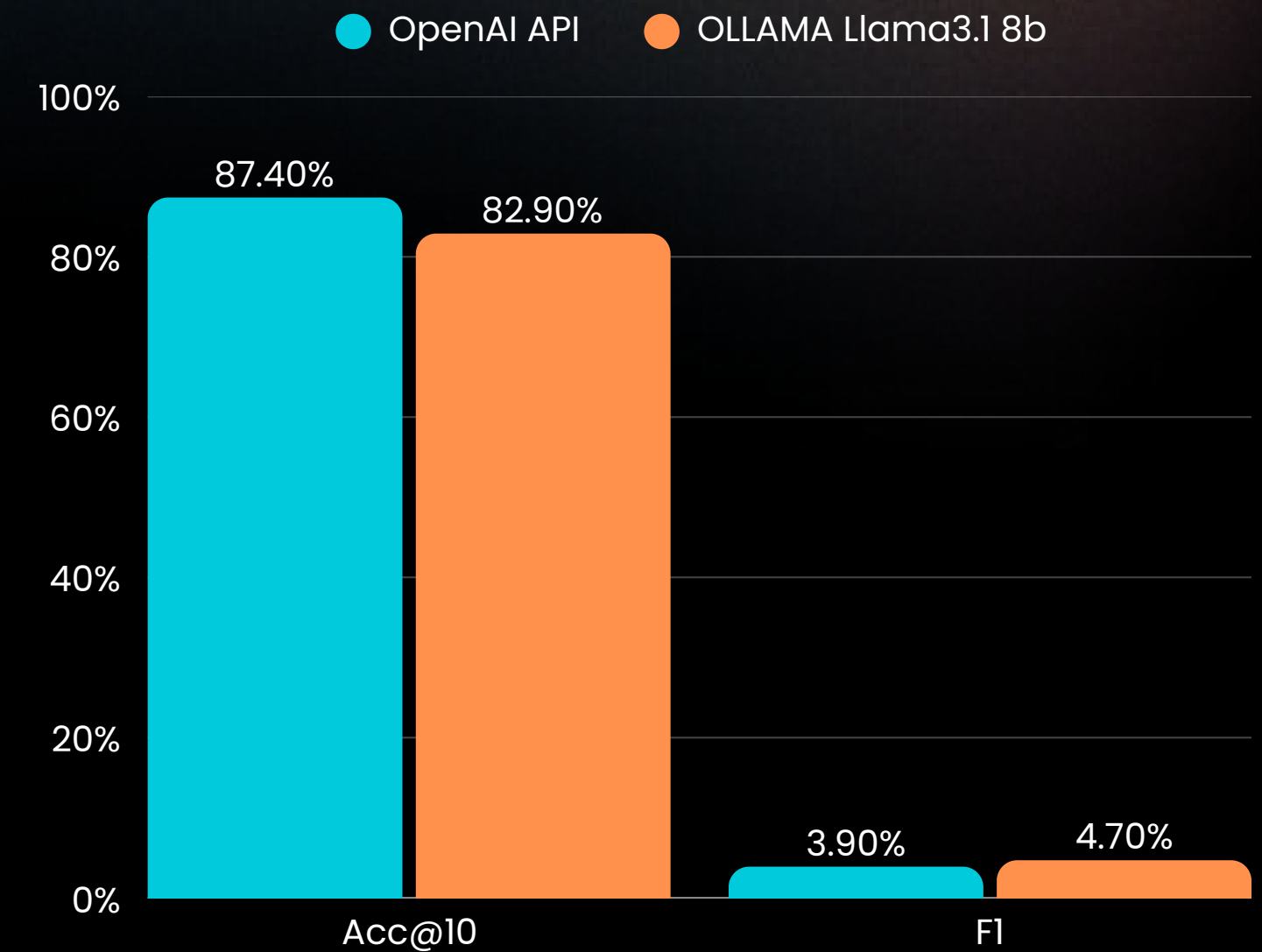
OpenAI API KEY

- A Pagamento
- No Privacy
- SOLO Modelli Di OpenAI



OLLAMA Llama3.1 8b

- Totalmente Gratuito
- Sicurezza Dei Dati
- TUTTI i Modelli Open Source



DATASET VERONACARD

03

ANAGRAFICA DEI POI (Point Of Interest)

- 22 Attrazioni Turstiche Di Verona
- Coordinate Geografiche
- Nomi Completi
- Nomi Abbreviati

LOG TRANSAZIONALI DEGLI ACCESSI

- Divisi Per Anni (2014 → 2023)
- TimeStamp Ingresso
- POI Visitato
- ID VeronaCard



STRATEGIE DI PROMPTING

04

APPROCCIO INCREMENTALE

BASE

NOME

- Prompt Semplice

GEOM

NOME
COORDINATE

- Prompt Medio

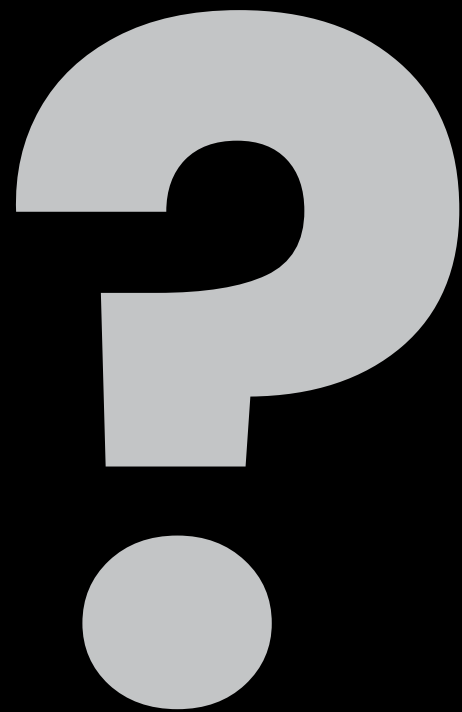
TIME

NOME
COORDINATE
TIMESTAMP

- Prompt Complesso

ANCHOR POINT

POI_1 —————> POI_2 —————> POI_3 —————> POI_4 —————> POI_5 —————> POI_6



ANCHOR POINT

01

POI_1 → POI_2 → POI_3 → POI_4 → POI_5 → POI_6

PENULTIMATE

 HISTORY

 CURRENT_POI

 PREDICTION

ANCHOR POINT

02

POI_1 → POI_2 → POI_3 → POI_4

MIDDLE

 HISTORY

 CURRENT_POI

 PREDICTION

AMBIENTE DI SVILUPPO

05

HPC LEONARDO

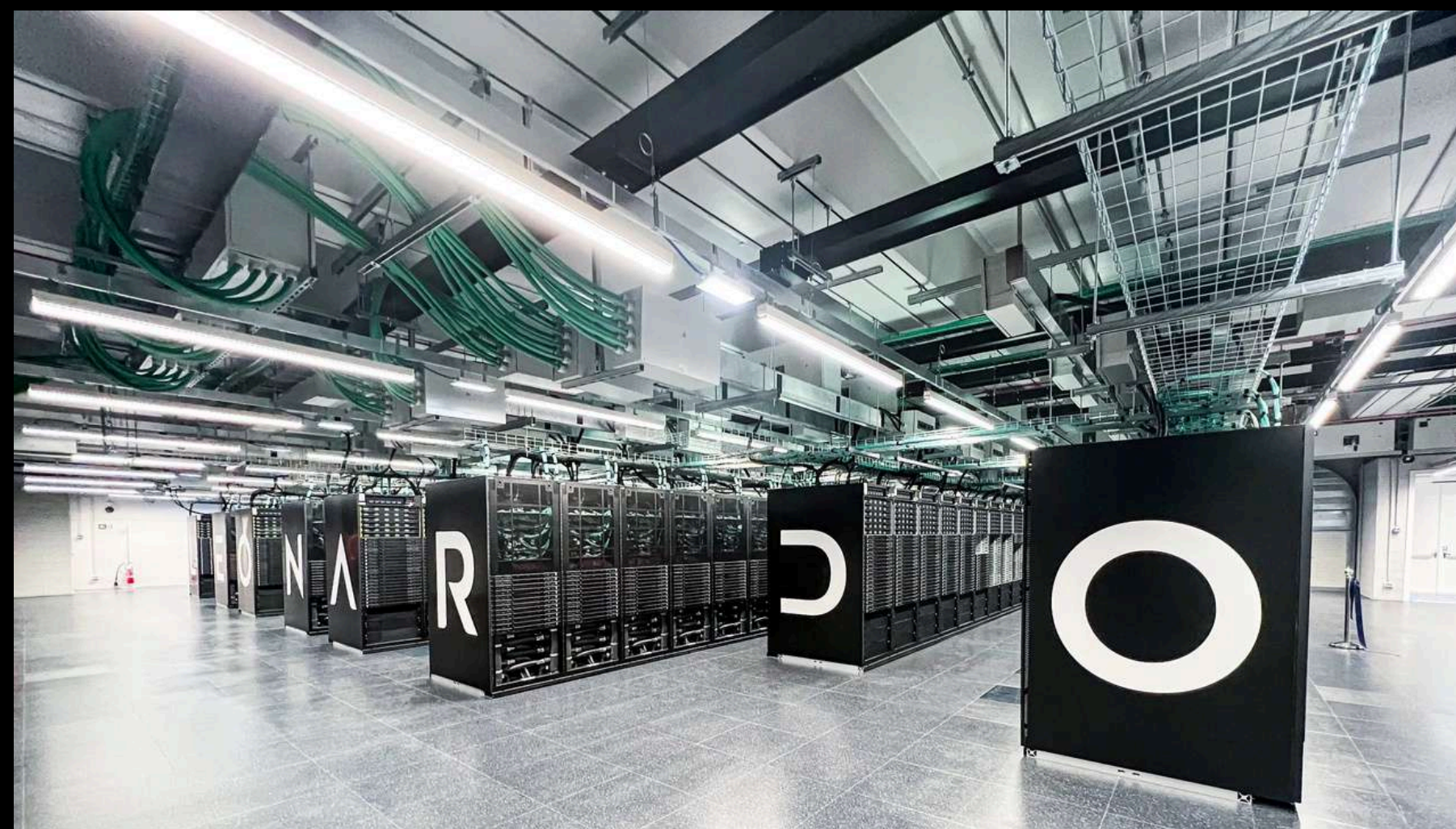
BOOSTER MODULE

Architettura

- 3456 NODI
- 512GB Di RAM DDR4 Per Nodo
- 4x NVIDIA Custom Ampere A100 Per Nodo

Configurazione Utilizzata

- Ollama Multi-Istanza su 4 GPU
- 12 Richieste Concorrenti (3 x GPU)
- Checkpoint System





DOCUMENTAZIONE

Studio per l'utilizzo corretto



SVILUPPO

Nuove funzionalità



DEVELOPMENT

Attesa allocazione risorse



ESECUZIONE

Su Leonardo

SPEED UP OTTENUTO

Grazie all'HPC Leonardo

PRIMA

- 2 Secondi Per Card
- Settimane Di Esecuzione

DOPO

- Da 7 a 12 Cards Al Secondo
- Meno Di 24h Per Esecuzione

MODELLI CONSIDERATI

06

LLAMA 3.1

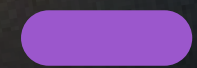
America

 **8B**

- Spazio Occupato → 4.9 GB
- RAM Necessaria → 8 GB
- Utilizzato per sviluppo e testing

QWEN 2.5

Cina



7B

- Spazio Occupato → 4.7 GB
- RAM Necessaria → 8 GB
- Ottima Comprensione Contestuale



14B

- Spazio Occupato → 9.0 GB
- RAM Necessaria → 16 GB
- Impatto Scaling

MISTRAL

Francia

MISTRAL 7B

- Spazio Occupato → 4.4 GB
- RAM Necessaria → 8 GB
- Molto Efficiente Su Singola GPU

MIXTRAL 8x7B

- Spazio Occupato → 26 GB
- RAM Necessaria → 26 GB
- Architettura MoE

DEEPSEEK

Cina

CODER 33B

- Spazio Occupato → 18 GB
- RAM Necessaria → 32 GB
- Modello Specializzato

RISULTATI OTTENUTI

07

STRUTTURA

CARD ID	CLUSTER	STORICO VISITE	POI ATTUALE	PREDIZIONE	GROUND TRUTH	MOTIVAZIONE	HIT	TEMPO PROC.	STATUS
0401314A216285	0	Santa Anastasia Castelvecchio	Arena	Casa Giulietta, Torre Lamberti, Museo Lapidario, Verona Tour, Tomba Giulietta	Casa Giulietta	Based on proximity and popularity among tourists visiting Arena, these POIs are likely to be the next destinations.	True	1.70s	Success

METRICHE DI VALUTAZIONE

$$\text{Top-1 Accuracy} = \frac{1}{N} \sum_{i=1}^N \begin{cases} 1 & \text{se } y_i = \hat{y}_{i,1} \\ 0 & \text{altrimenti} \end{cases}$$

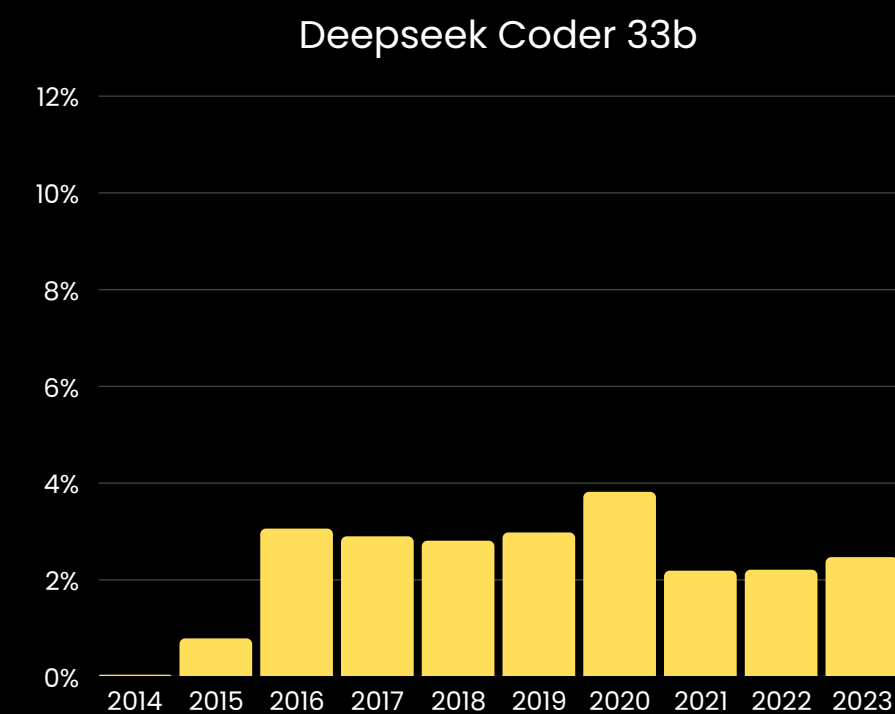
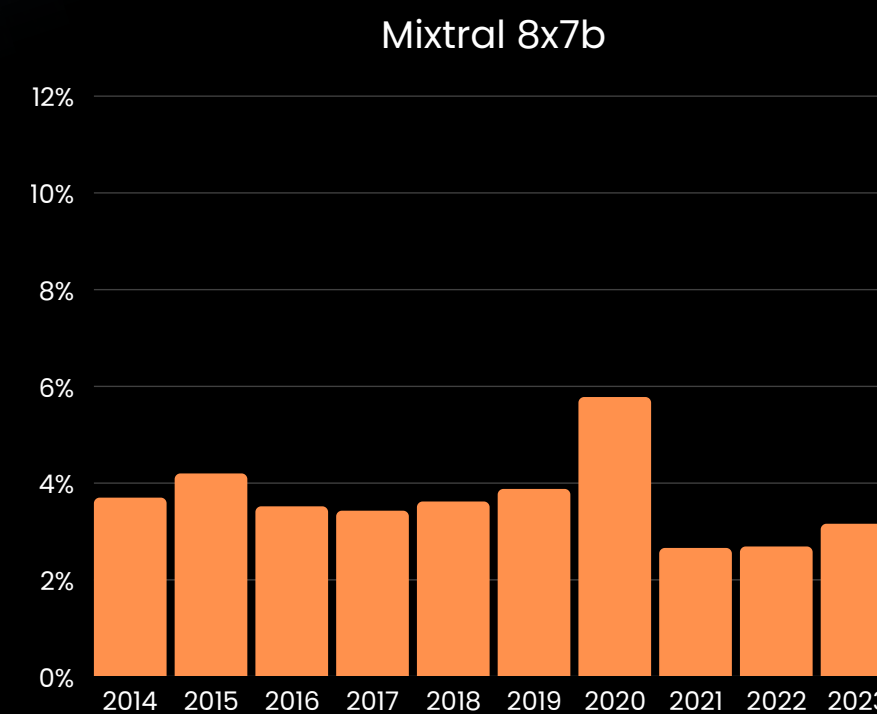
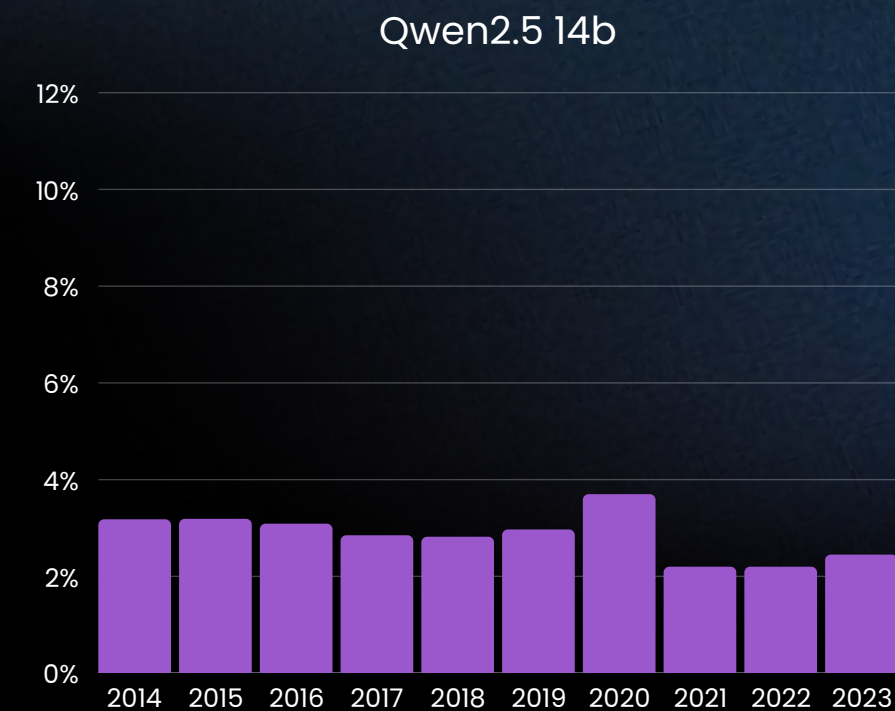
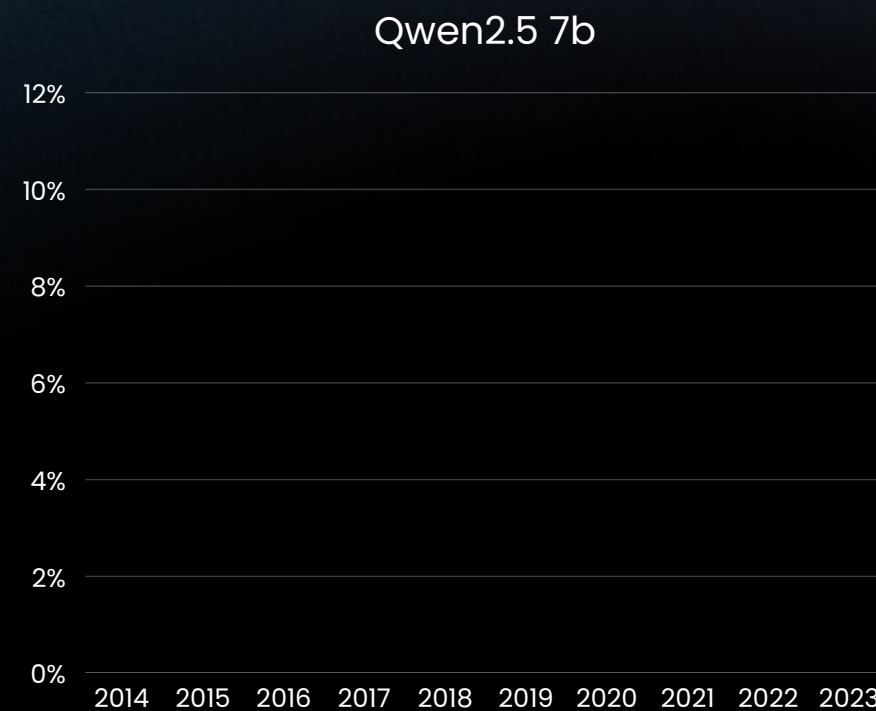
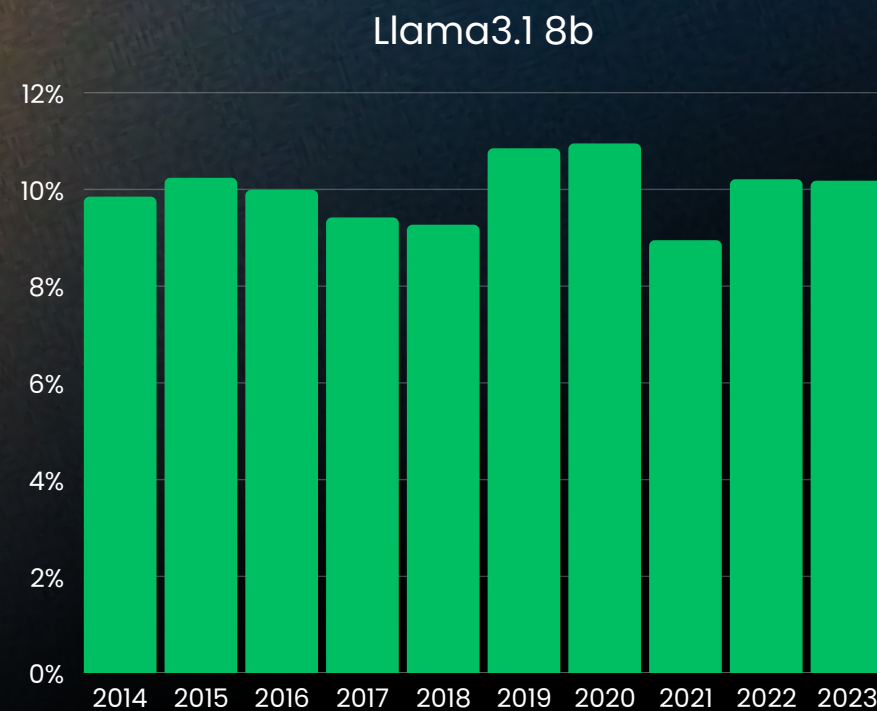
$$\text{Top-5 Hit Rate} = \frac{1}{N} \sum_{i=1}^N \begin{cases} 1 & \text{se } y_i \in \{\hat{y}_{i,1}, \dots, \hat{y}_{i,5}\} \\ 0 & \text{altrimenti} \end{cases}$$



N = Numero totale di predizioni y_i = POI realmente visitato $\hat{y}_{i,k}$ = POI predetto in posizione k

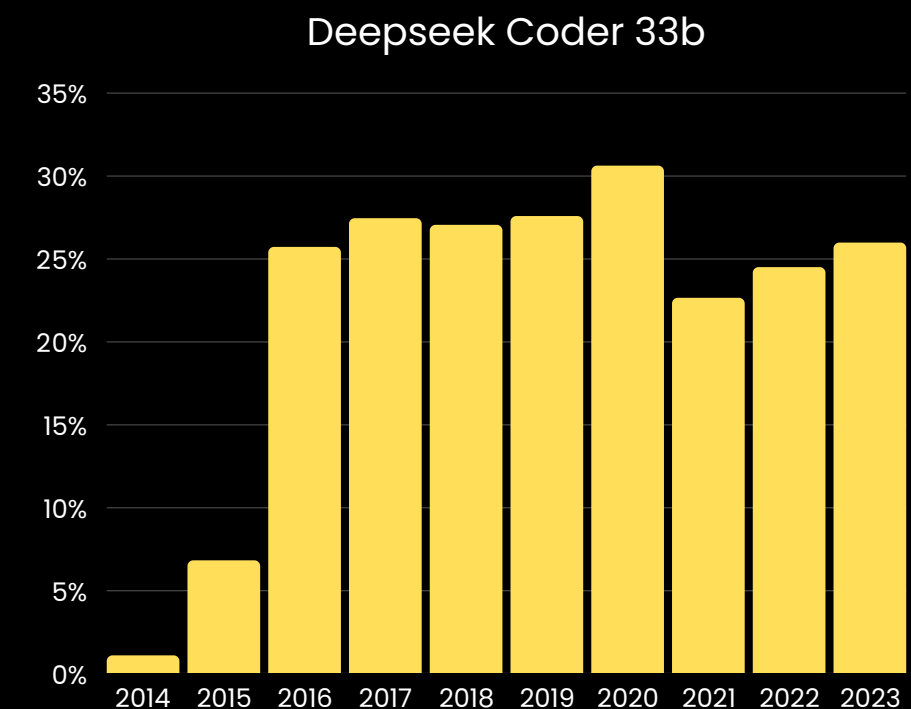
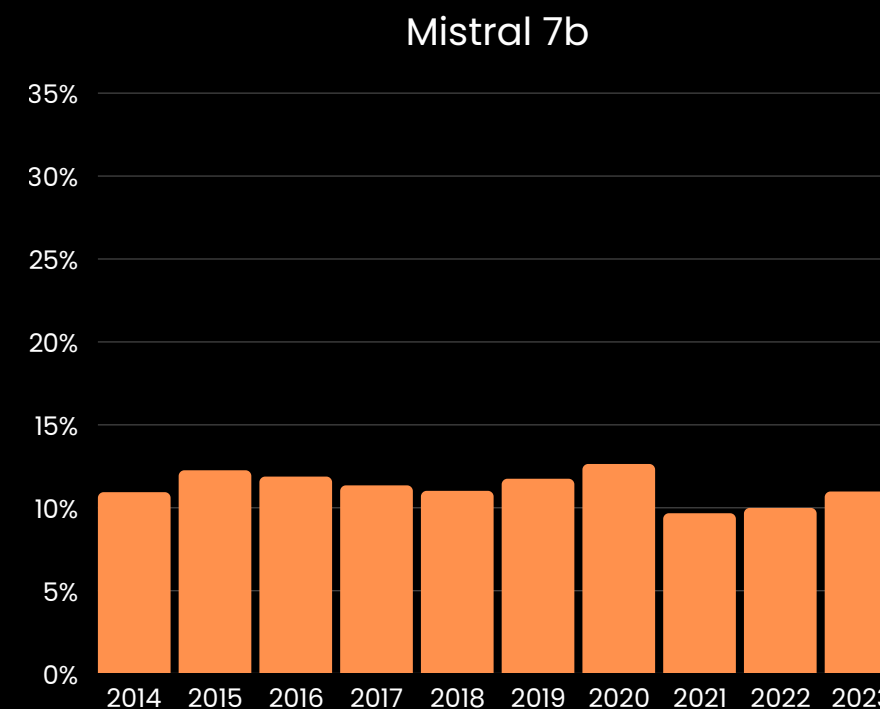
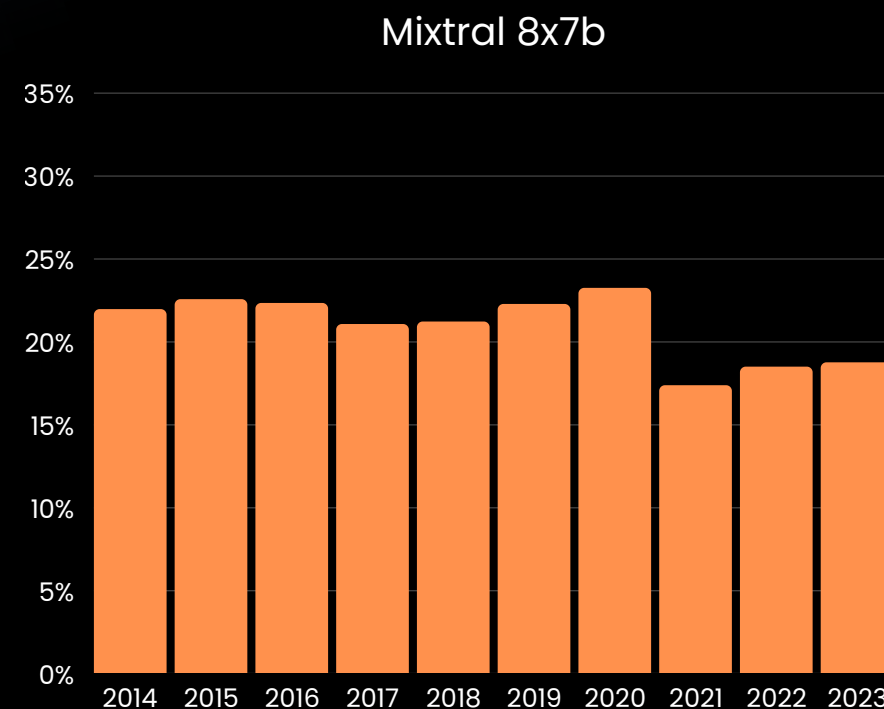
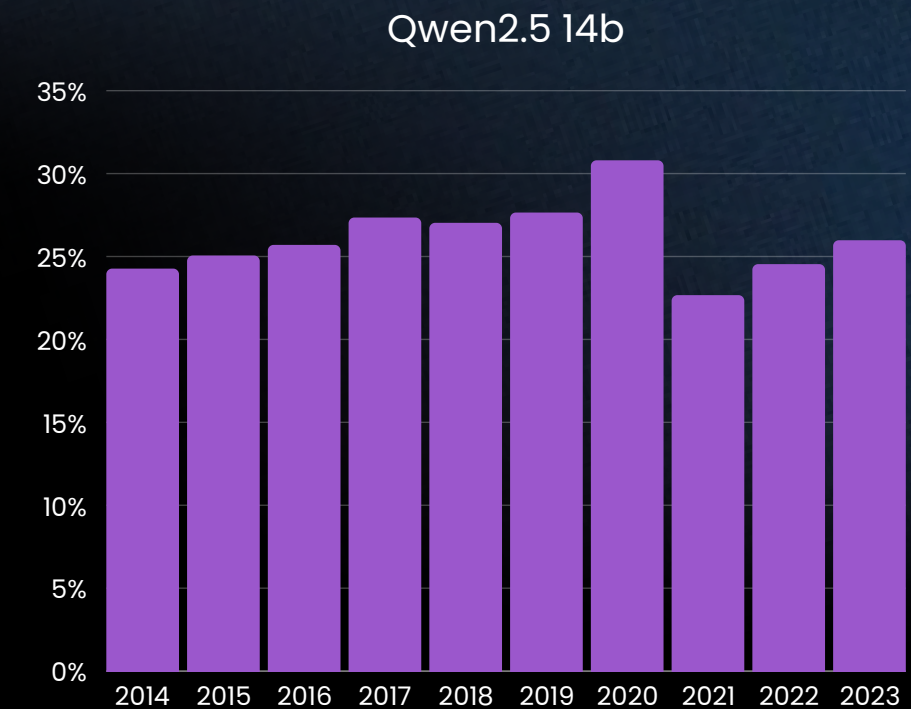
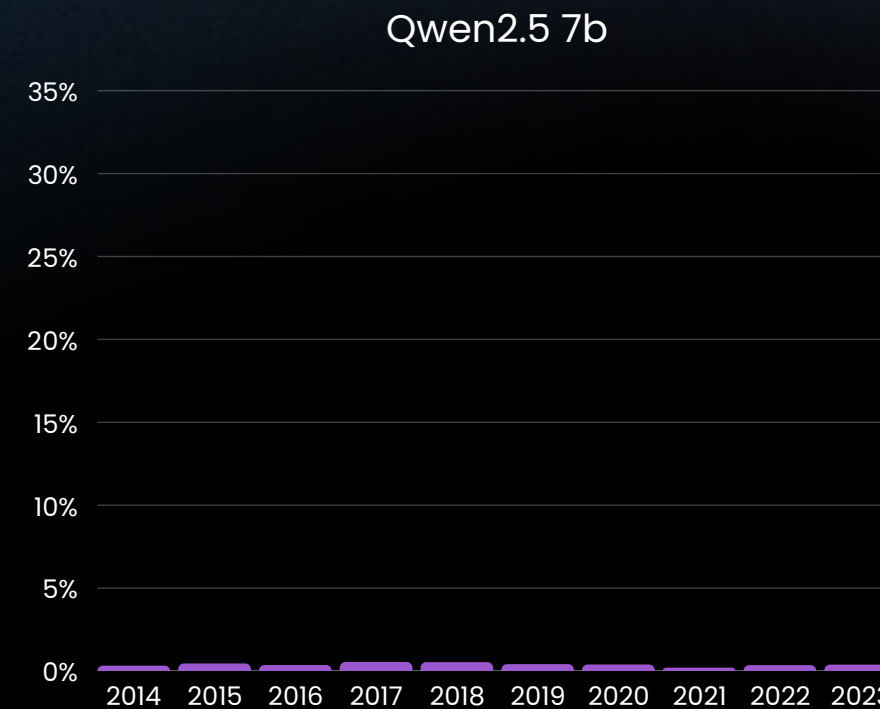
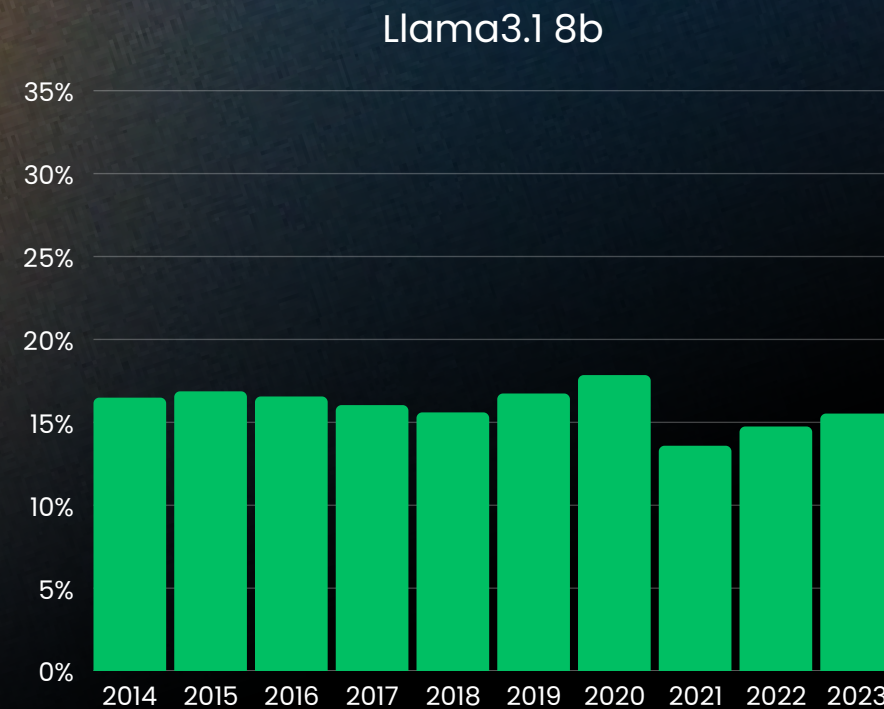
STRATEGIA BASE – PENULTIMATE

TOP 1 ACCURACY



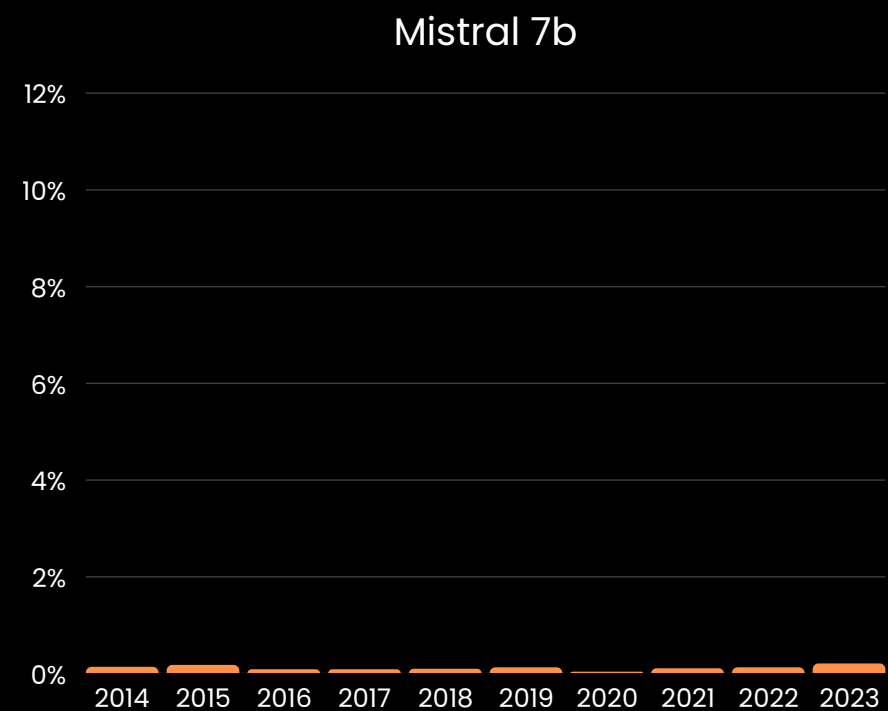
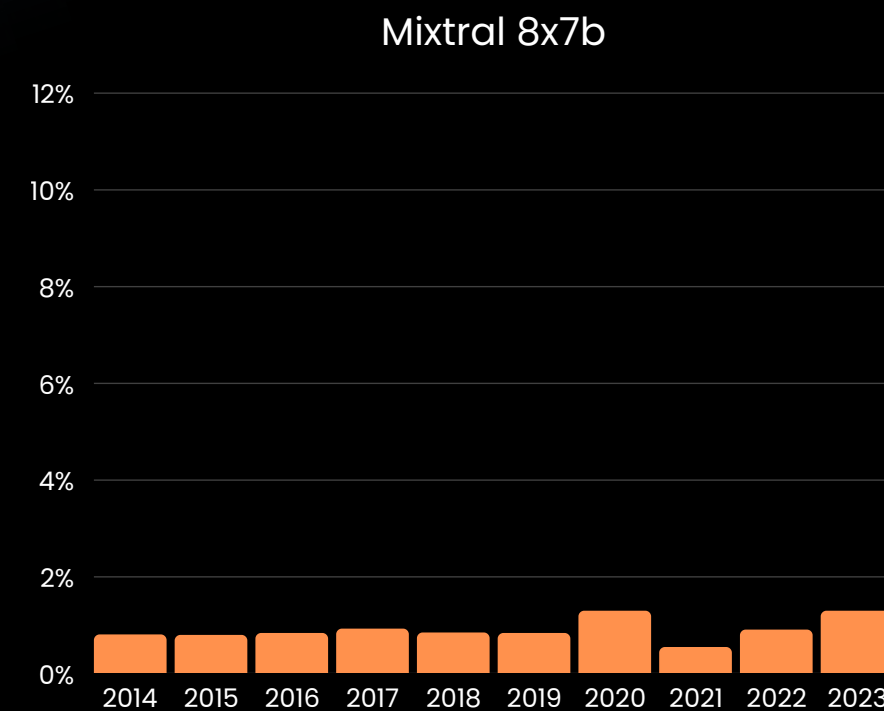
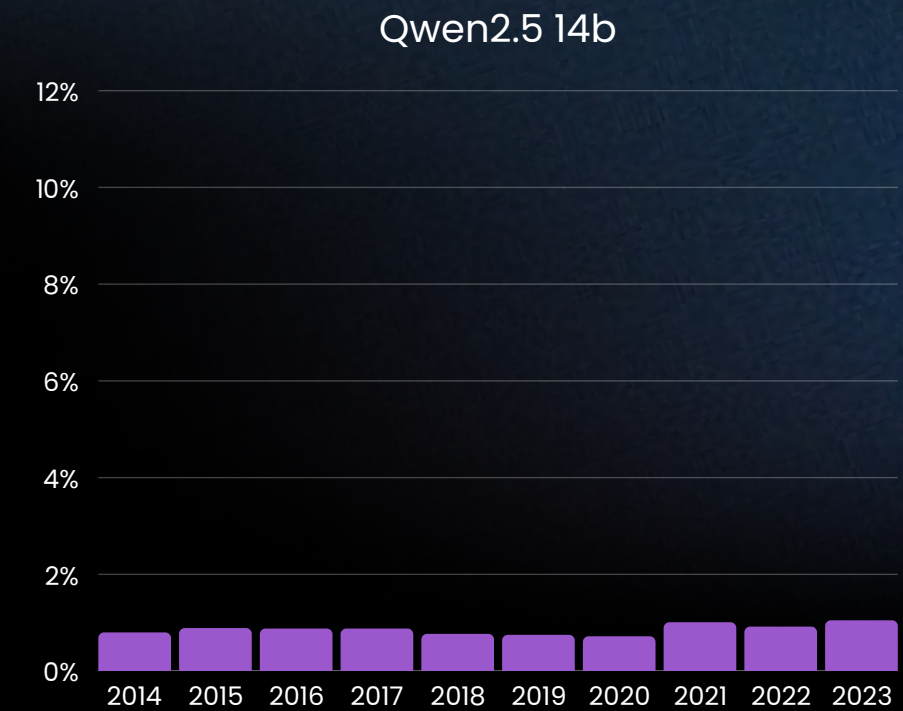
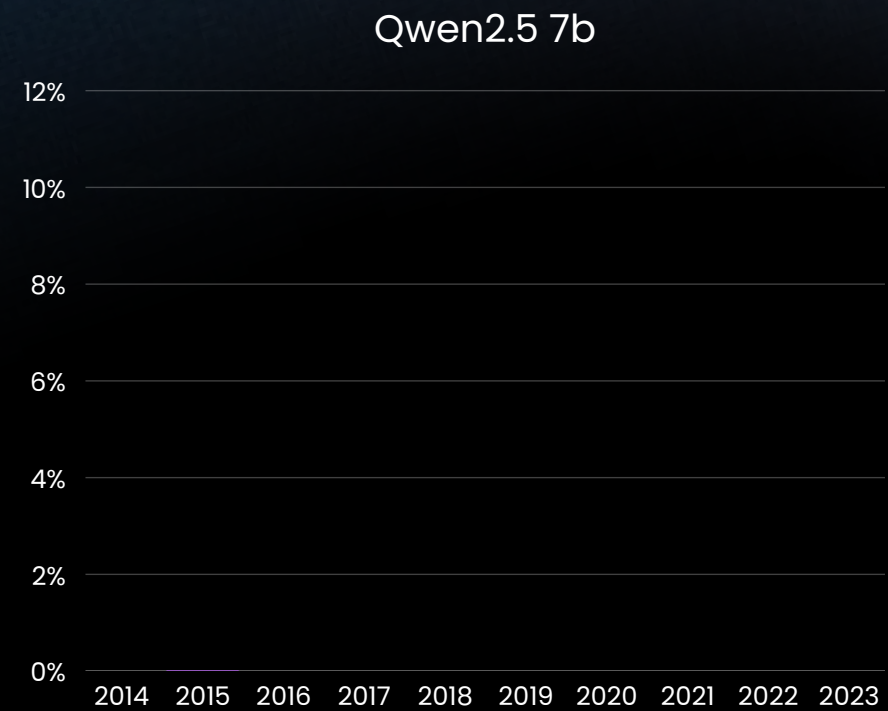
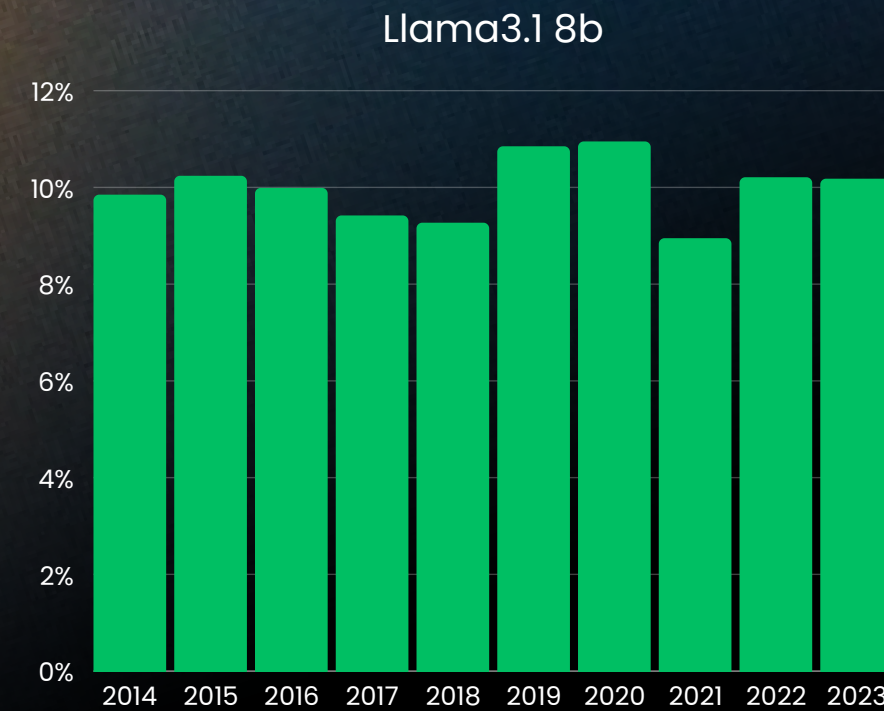
STRATEGIA BASE – PENULTIMATE

TOP-5 HIT RATE



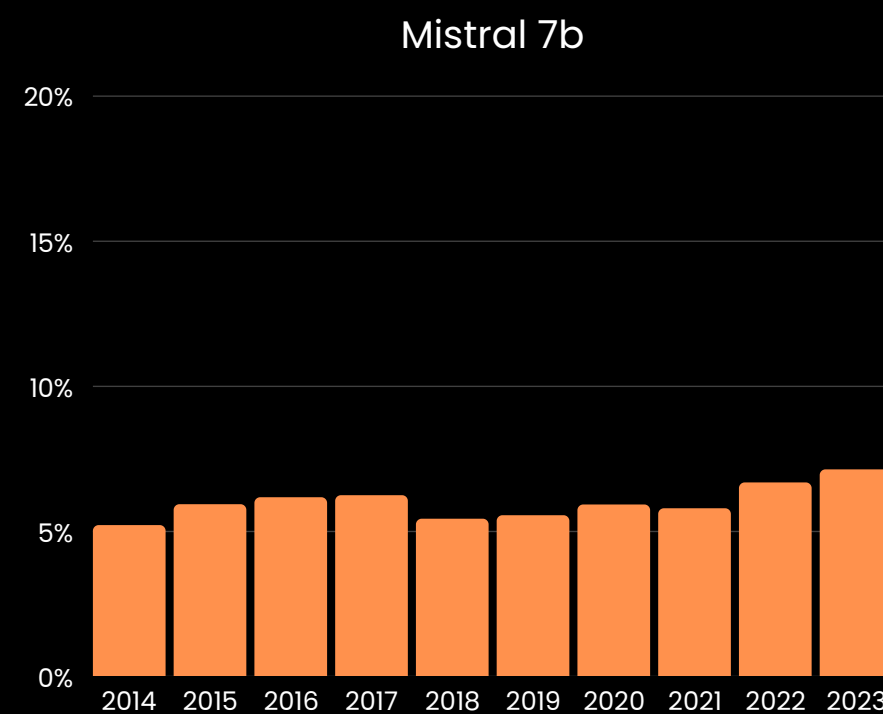
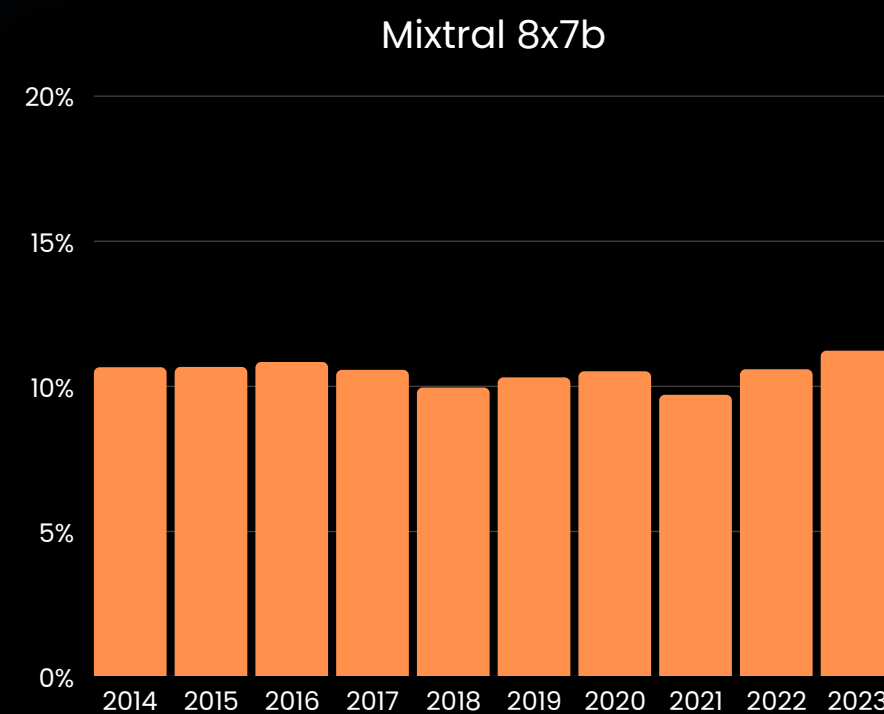
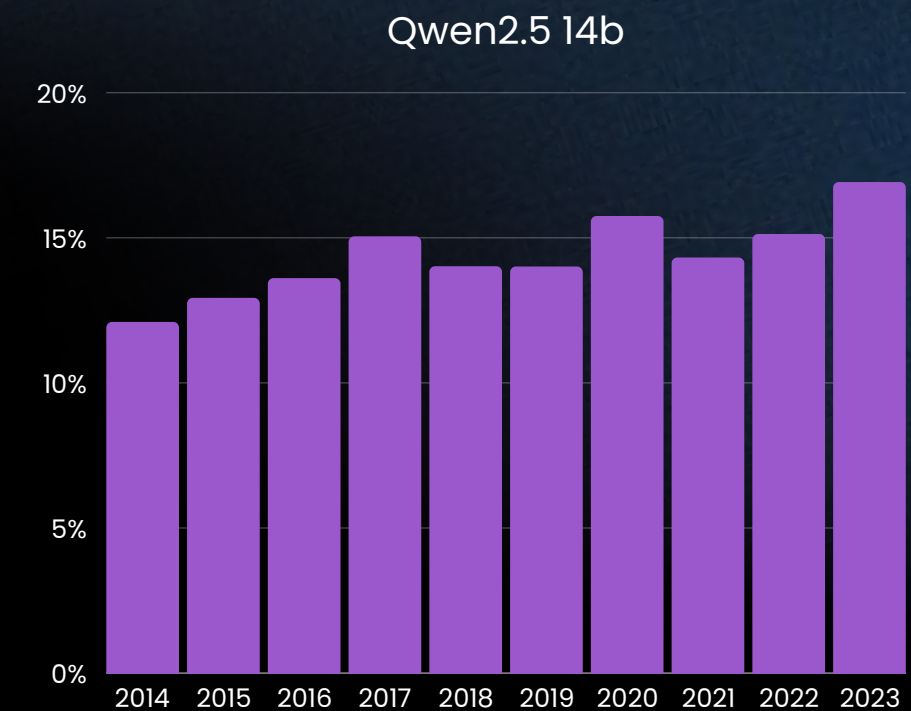
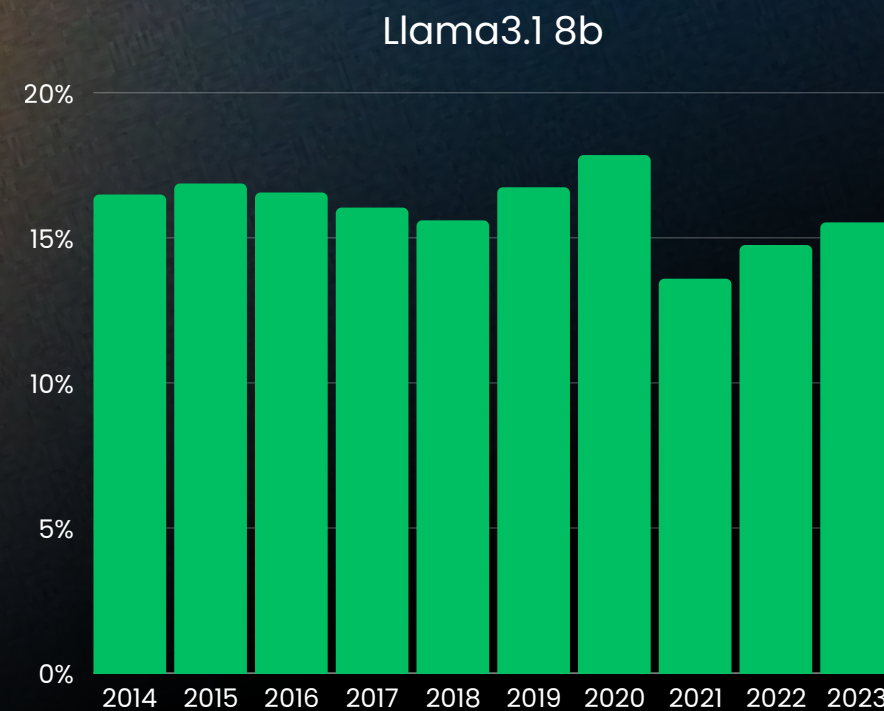
STRATEGIA BASE – MIDDLE

TOP 1 ACCURACY



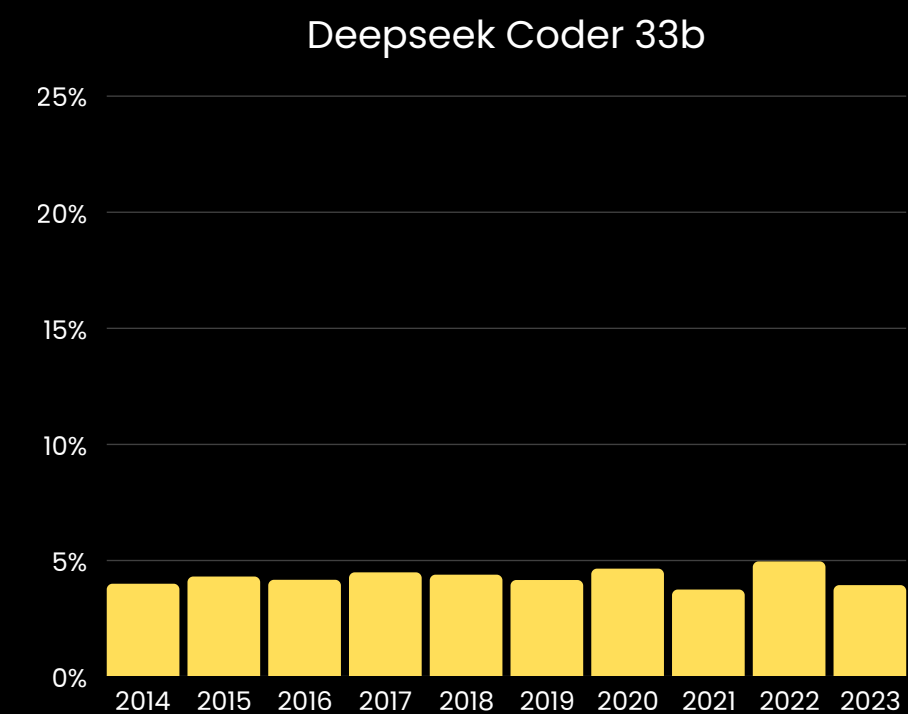
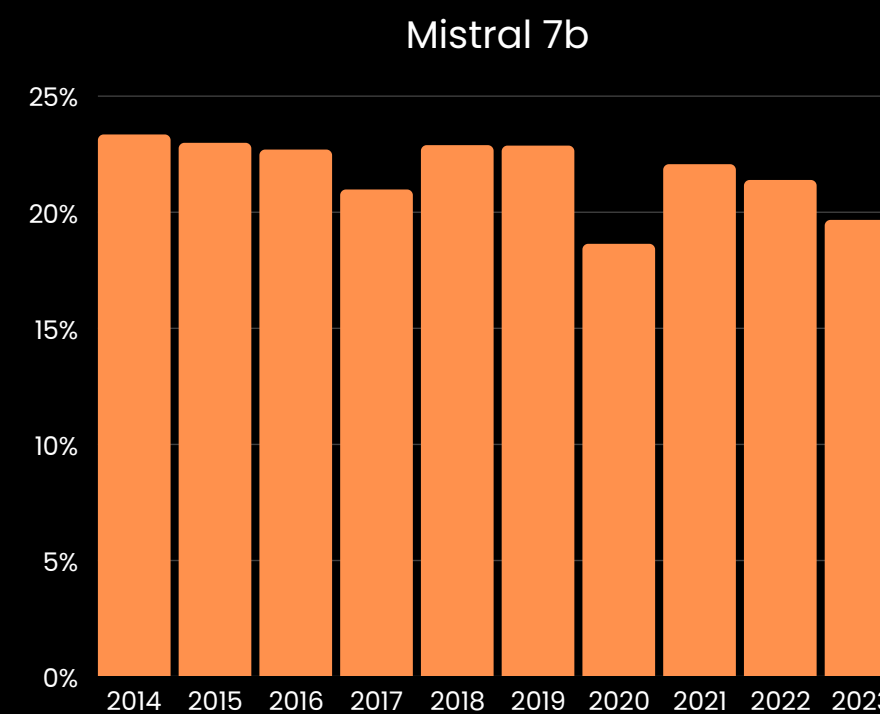
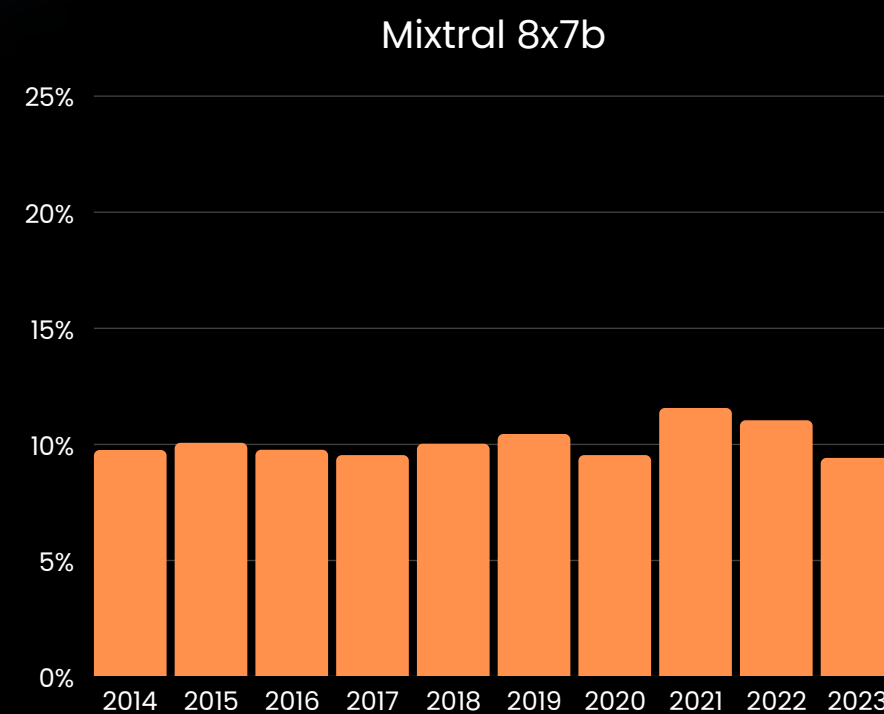
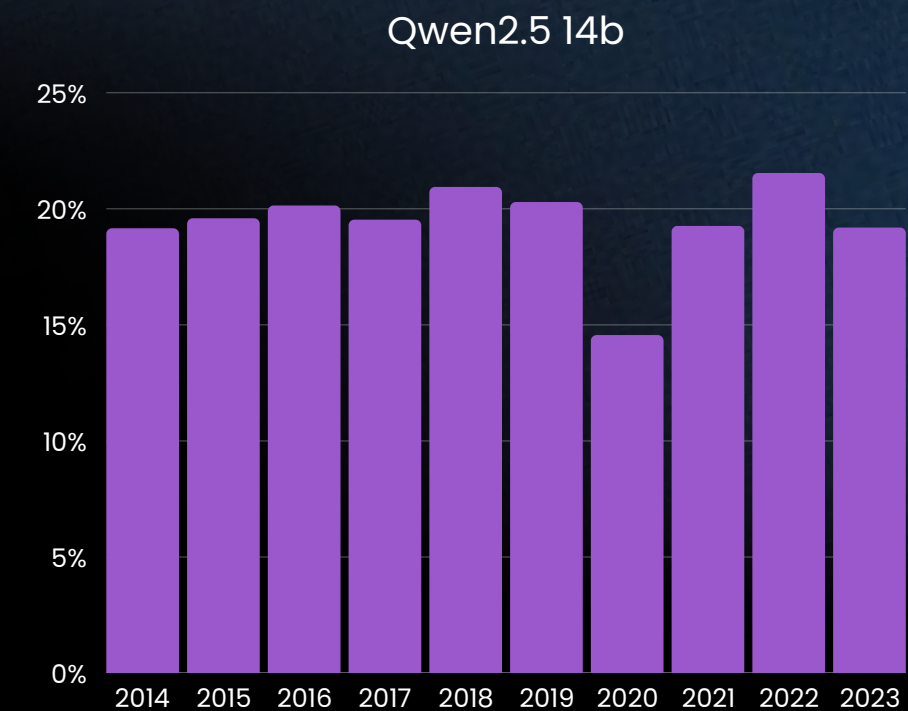
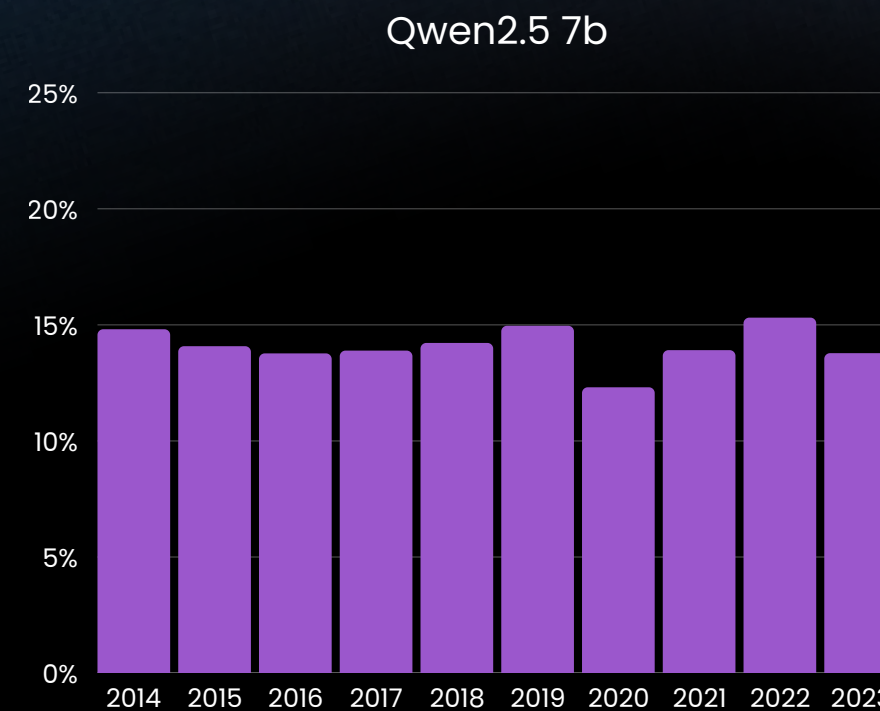
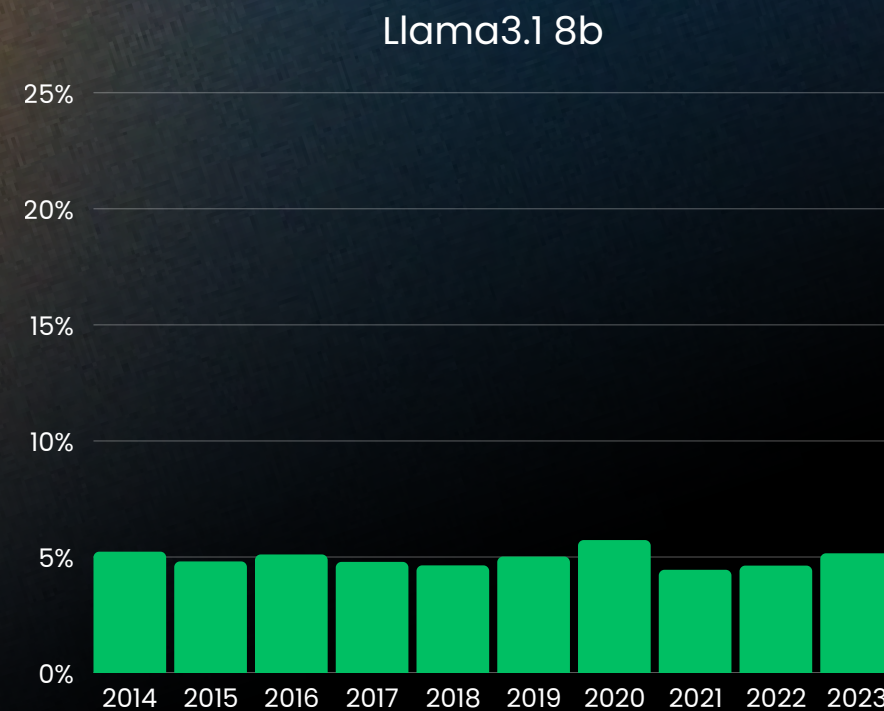
STRATEGIA BASE – MIDDLE

TOP-5 HIT RATE



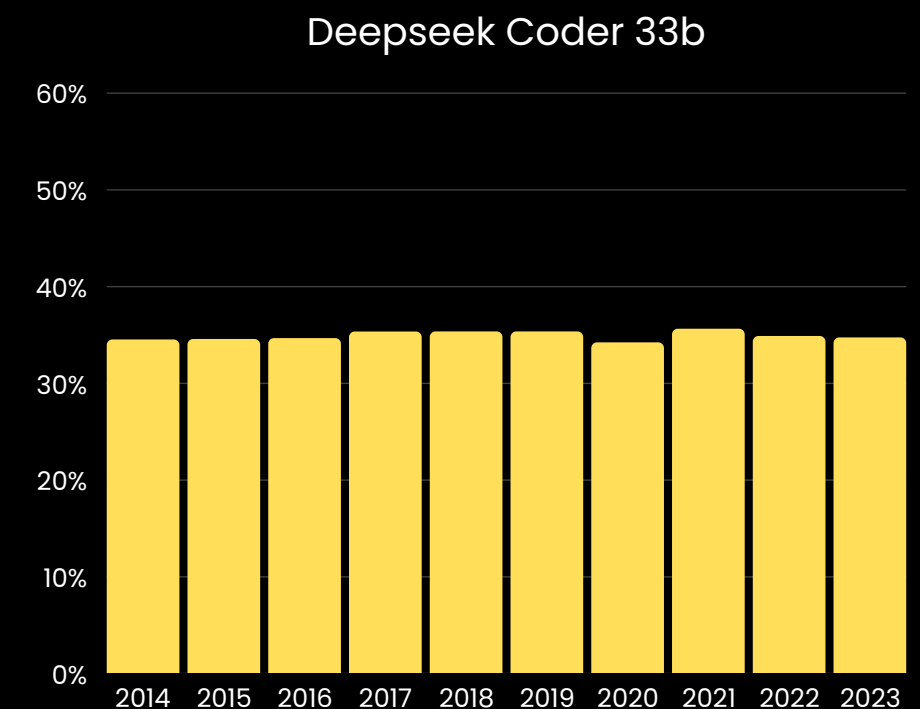
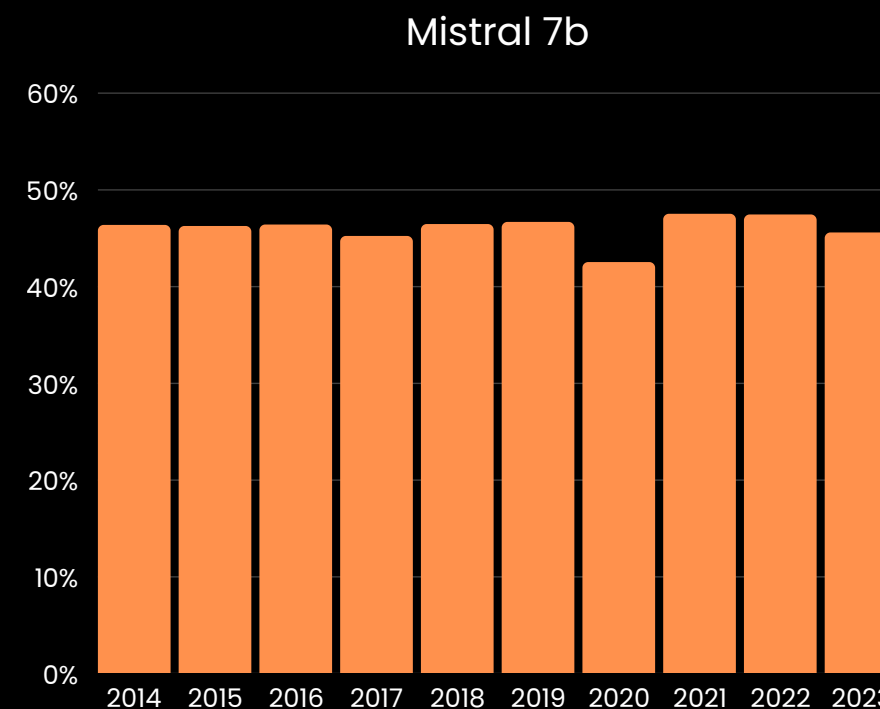
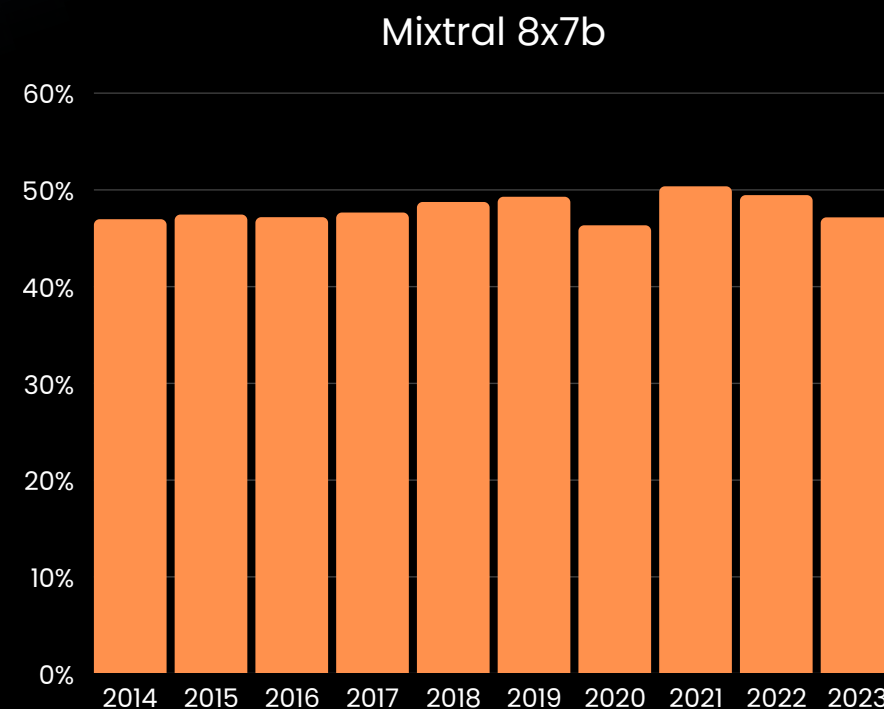
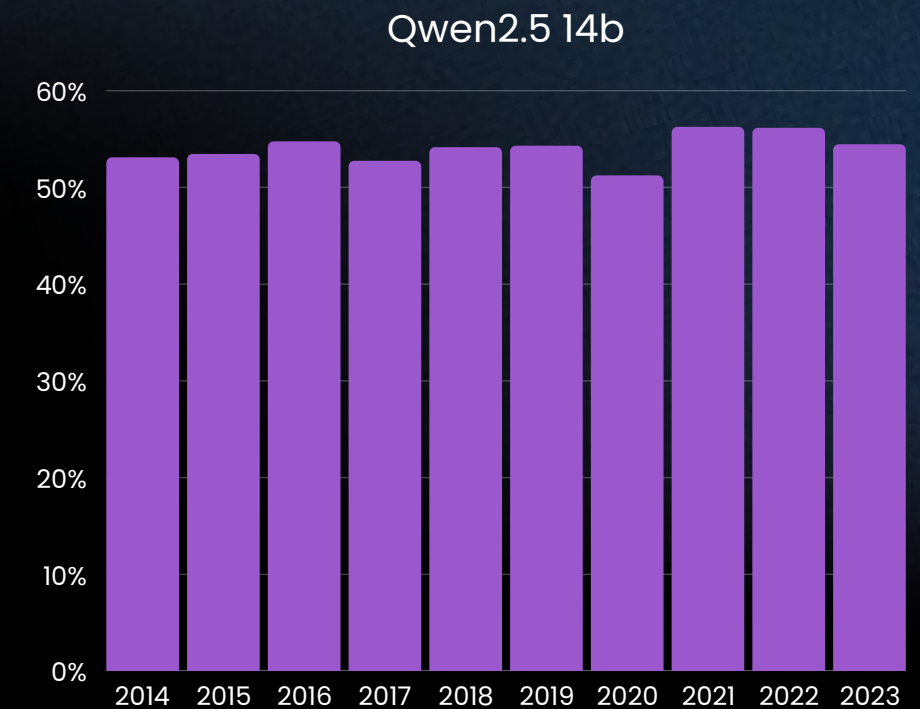
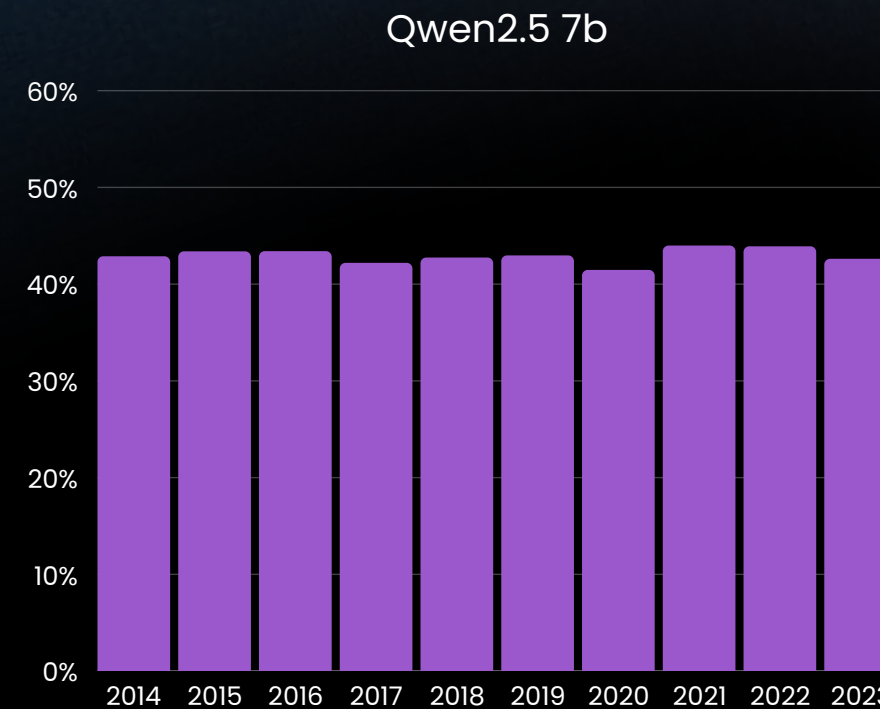
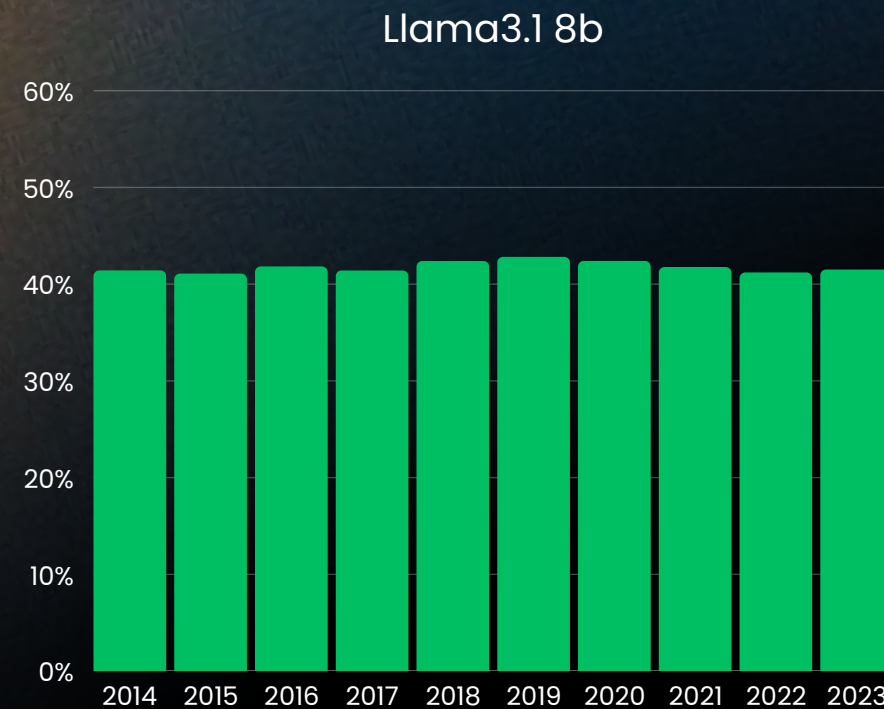
STRATEGIA GEOM – PENULTIMATE

TOP 1 ACCURACY



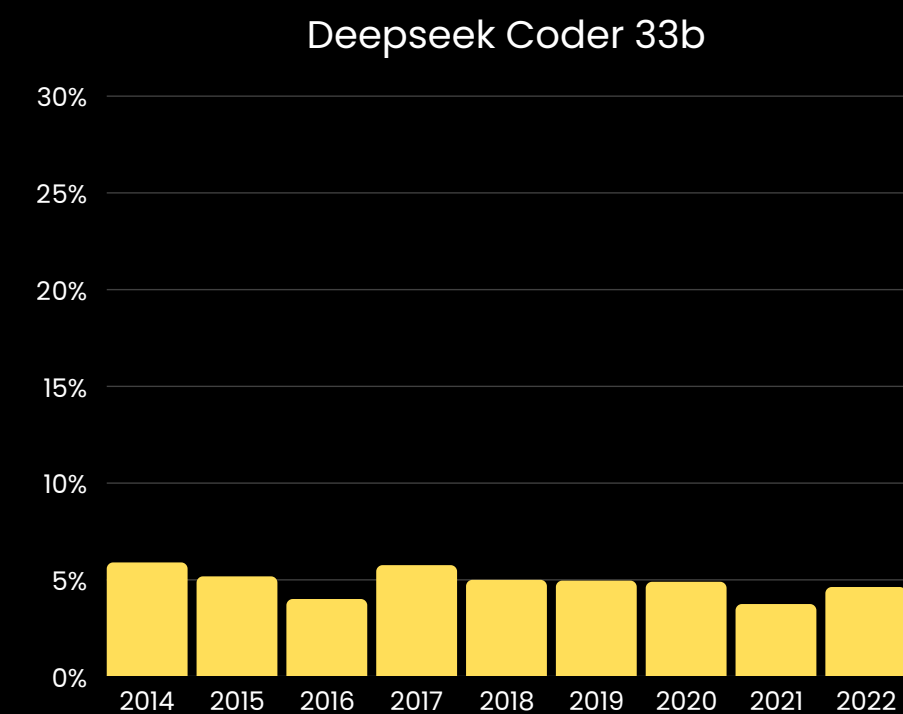
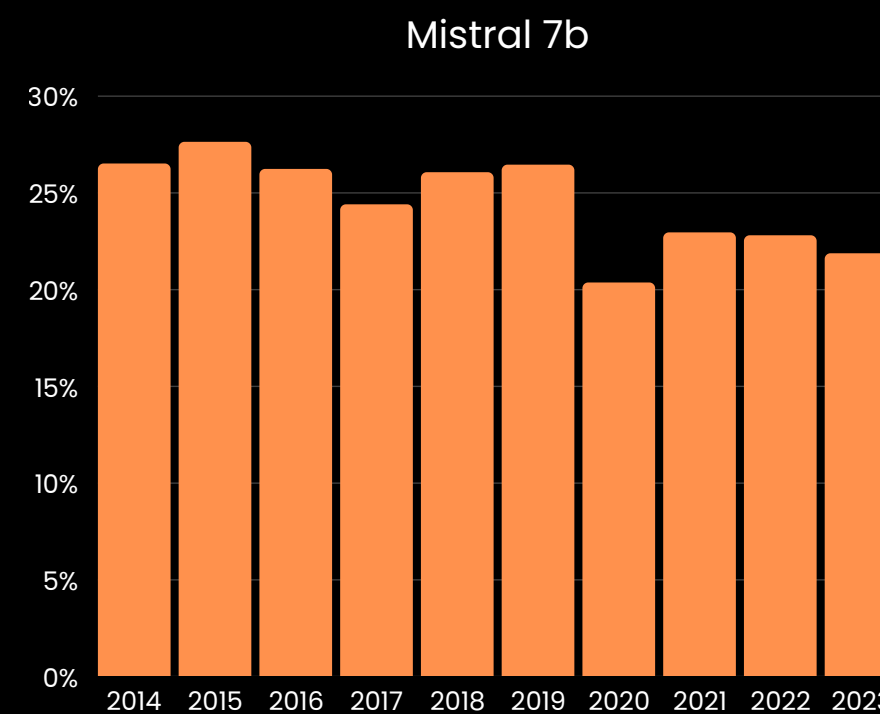
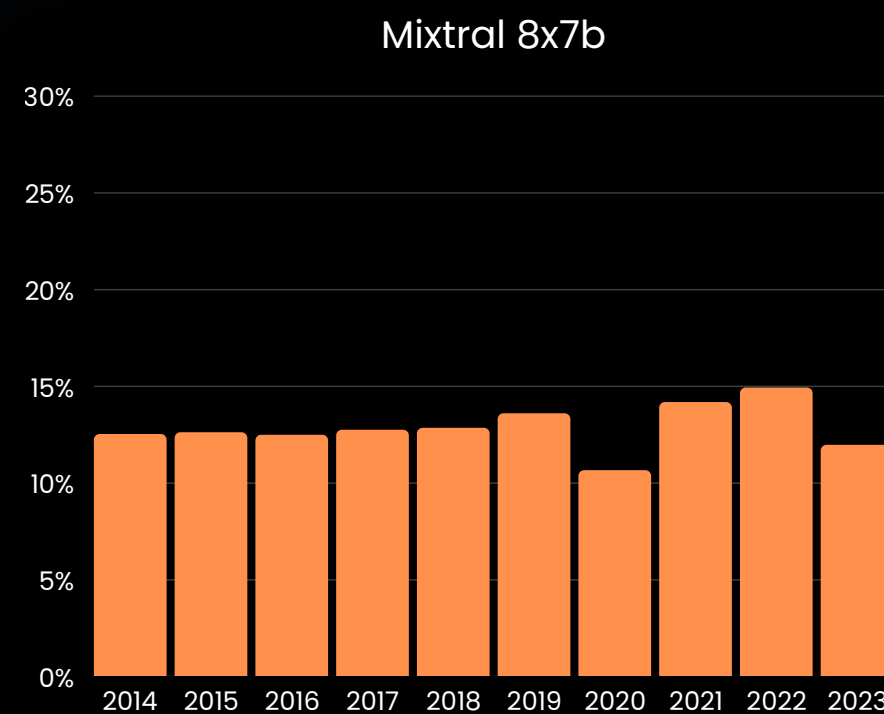
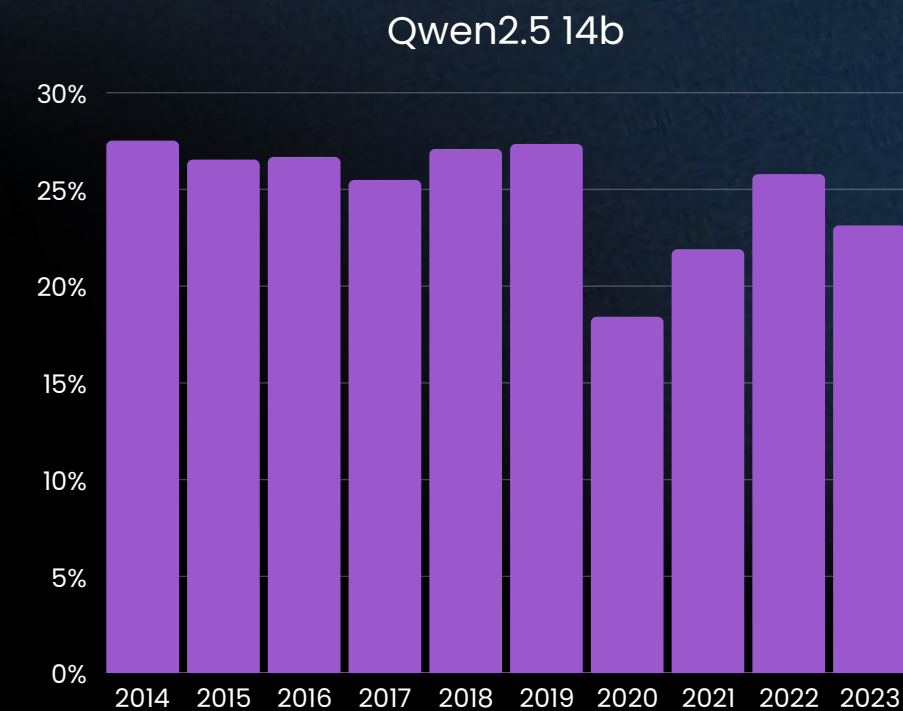
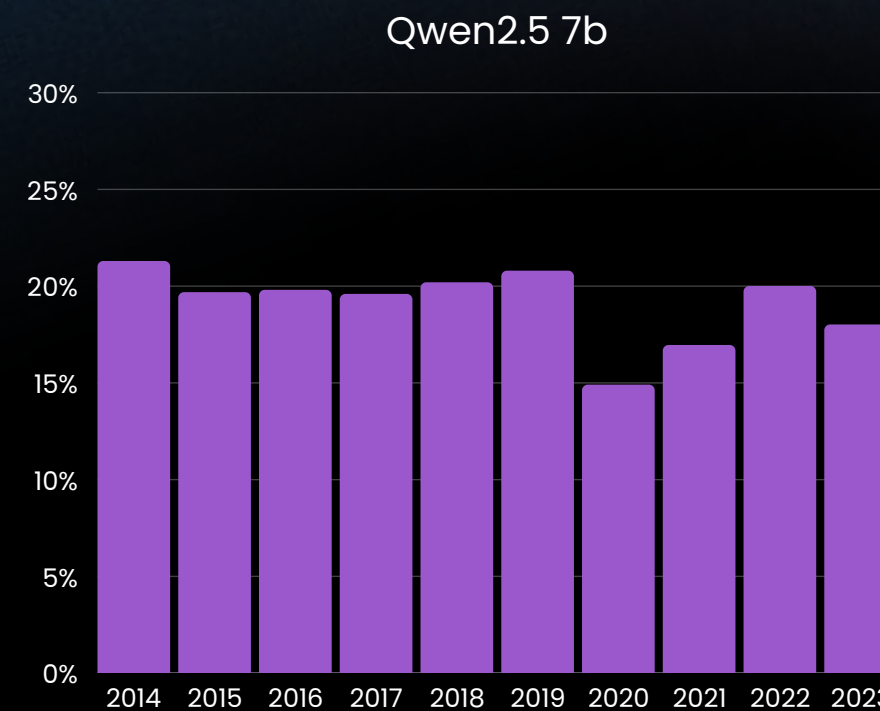
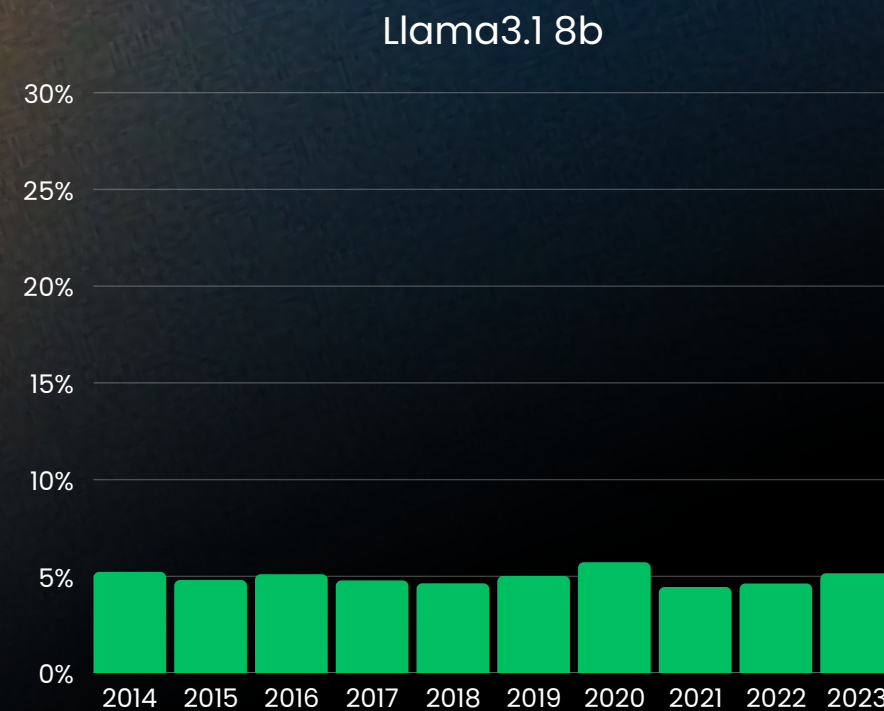
STRATEGIA GEOM – PENULTIMATE

TOP-5 HIT RATE



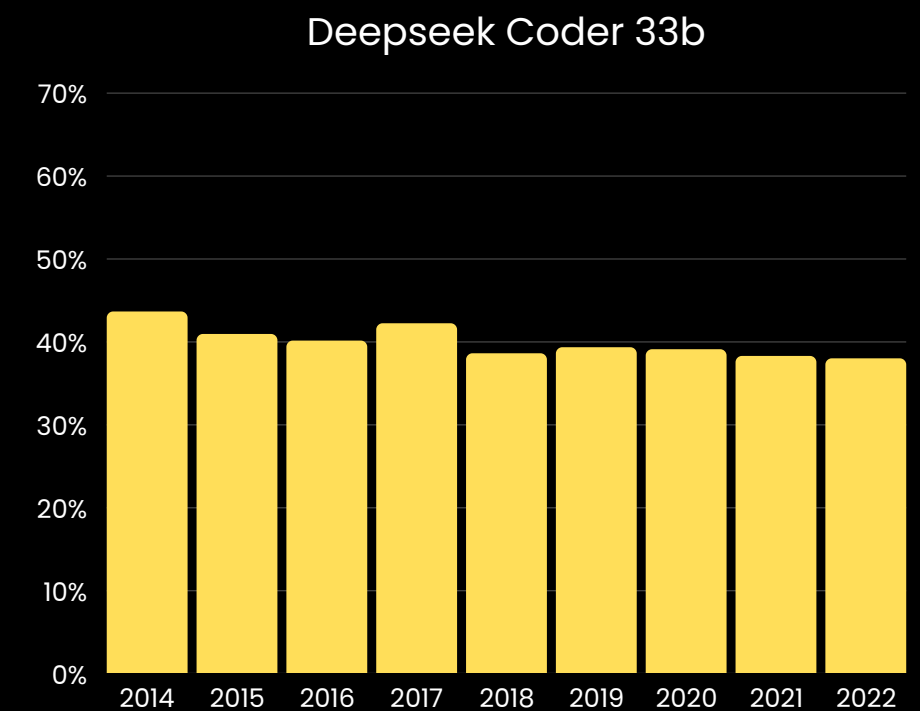
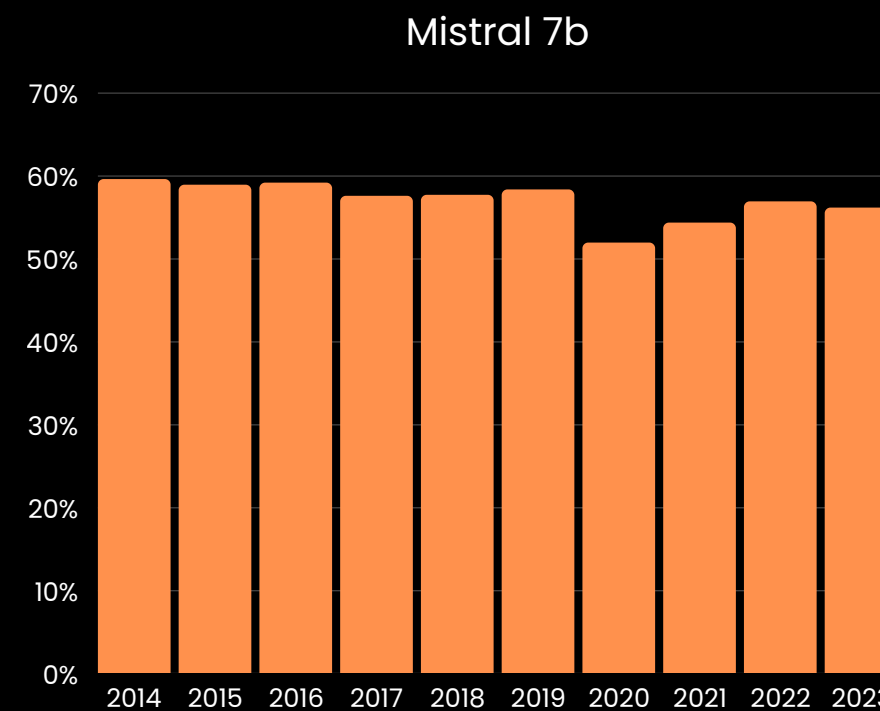
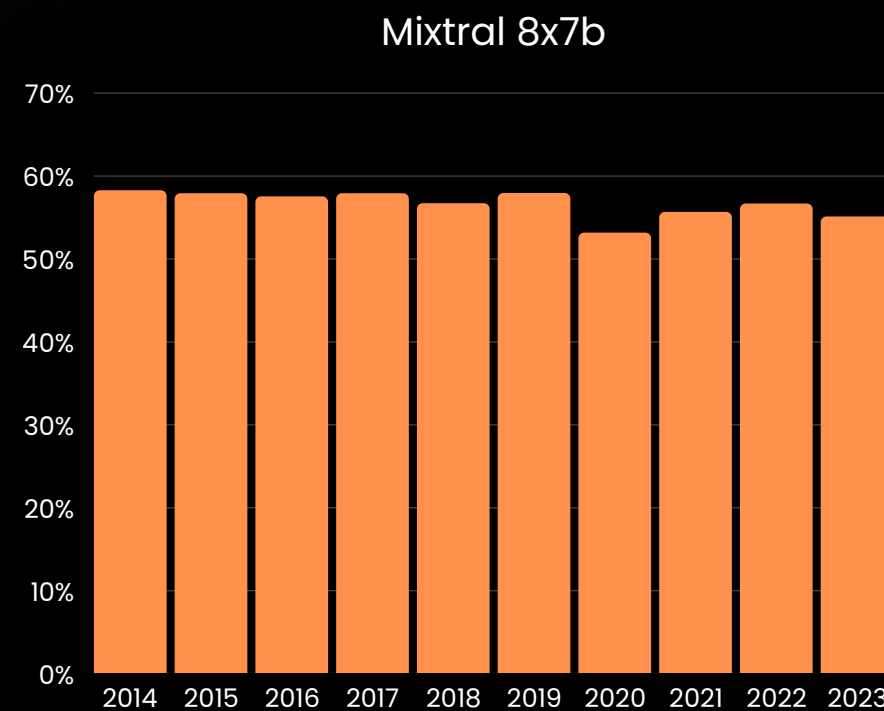
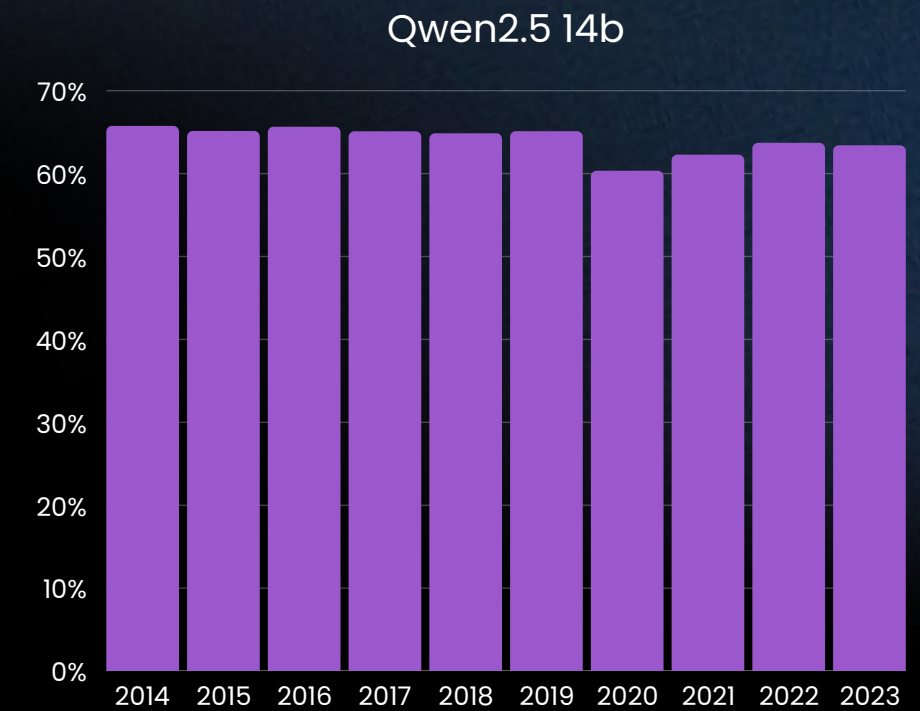
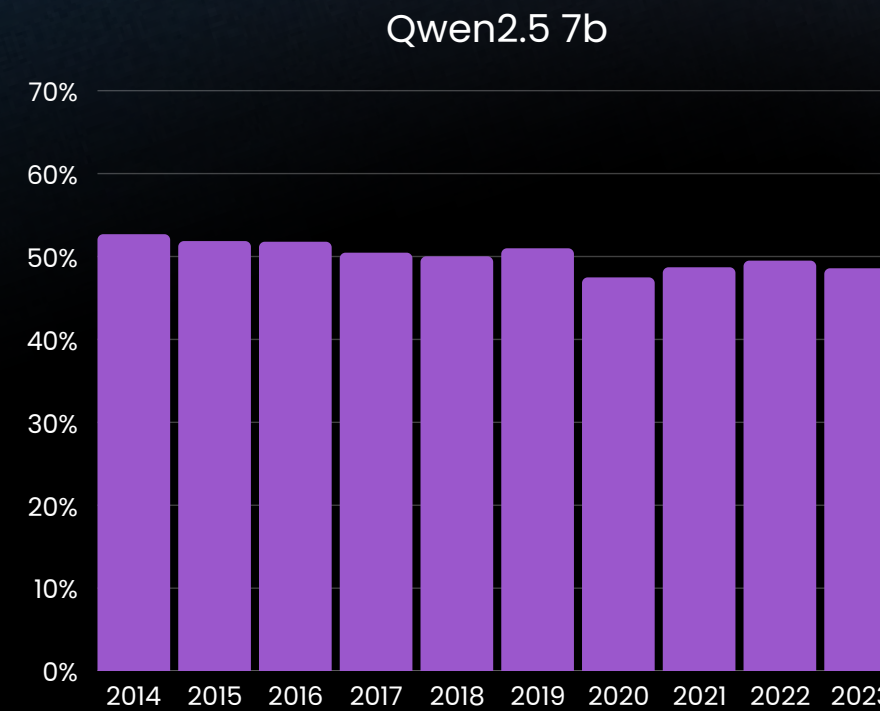
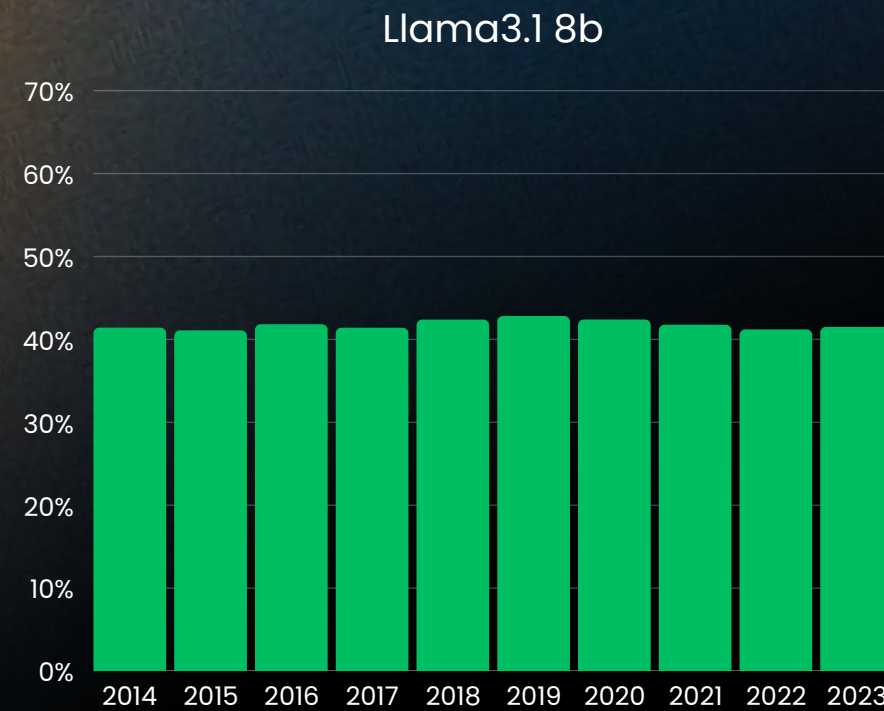
STRATEGIA GEOM – MIDDLE

TOP 1 ACCURACY



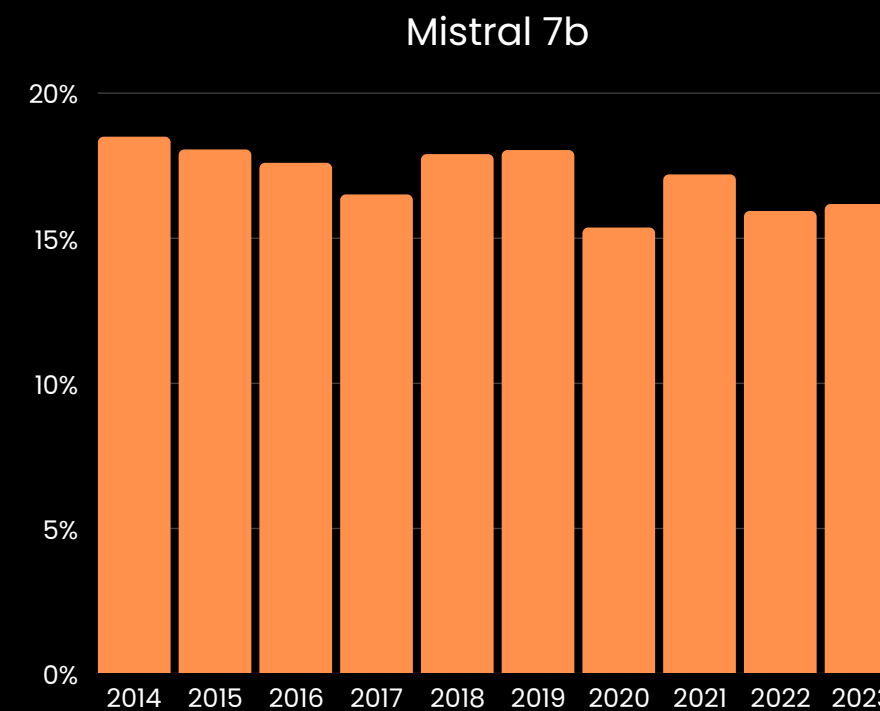
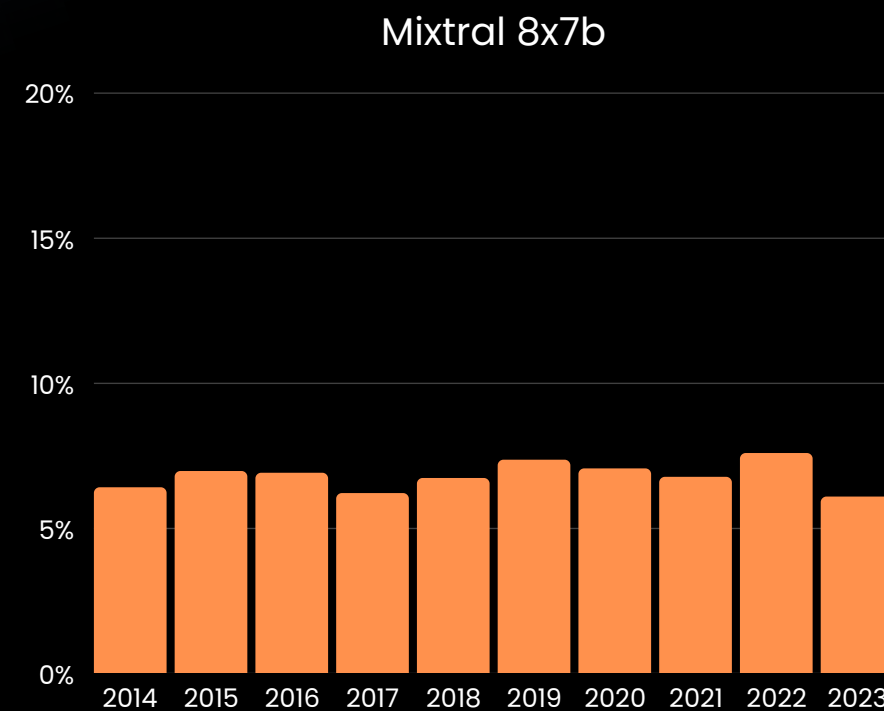
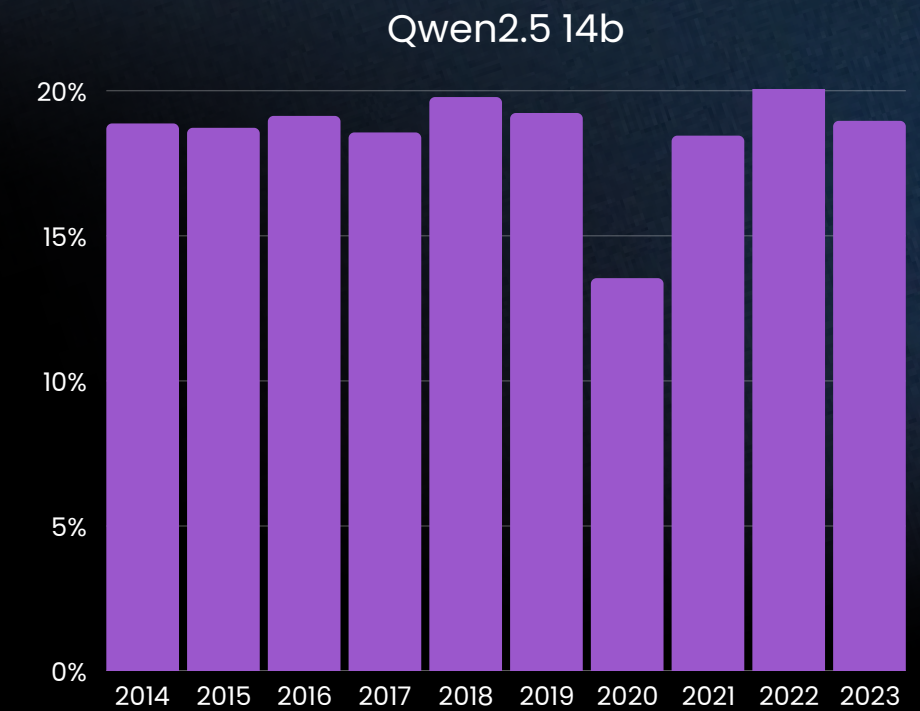
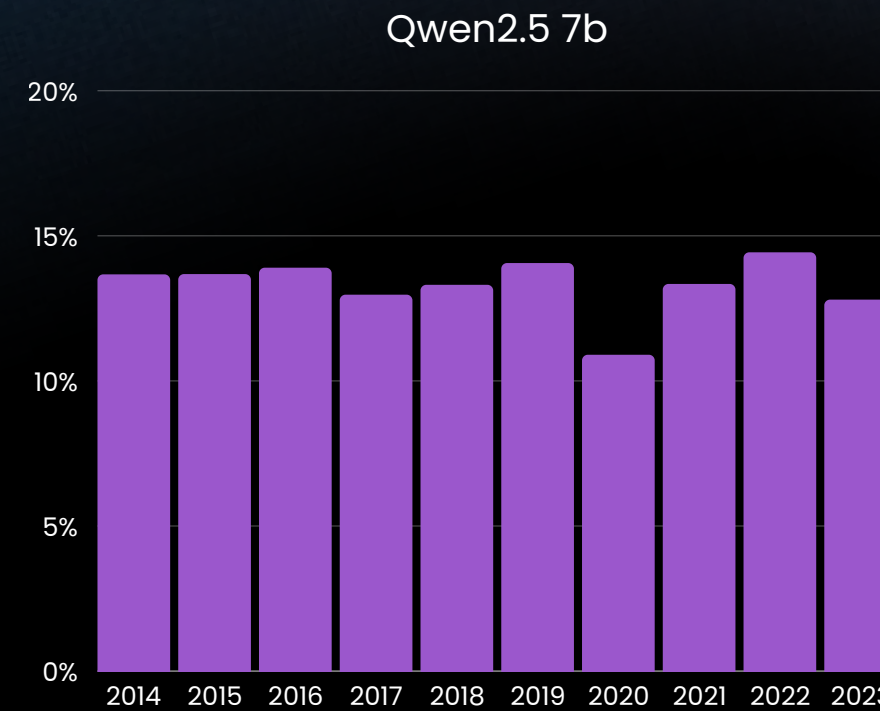
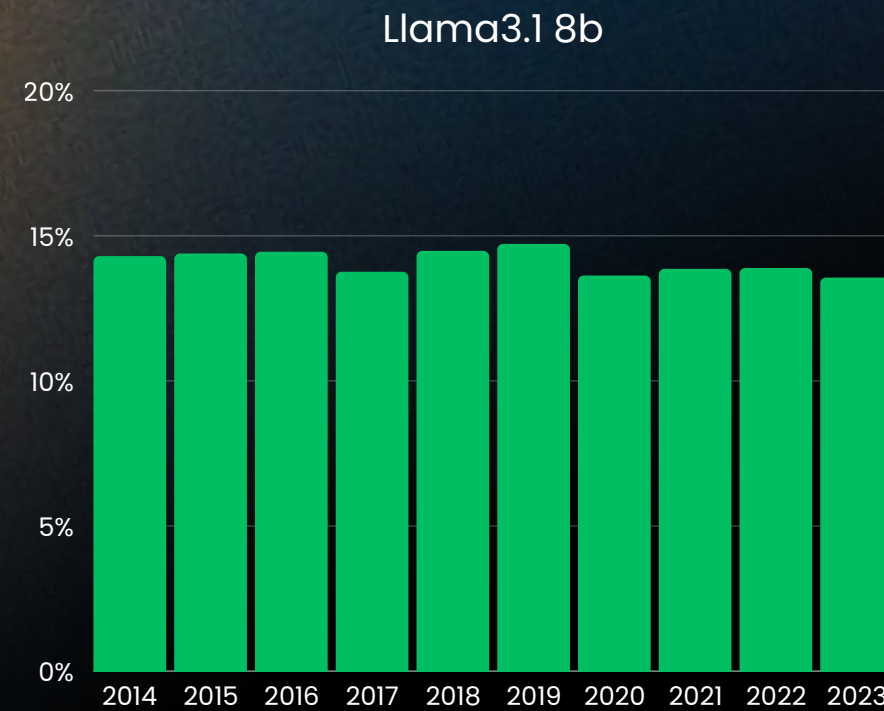
STRATEGIA GEOM – MIDDLE

TOP-5 HIT RATE



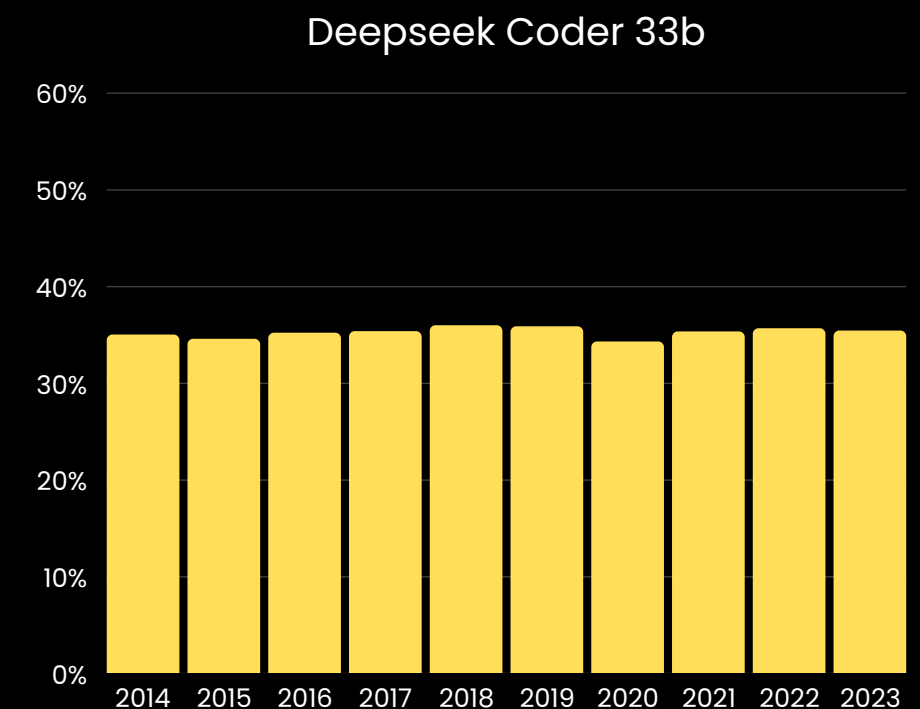
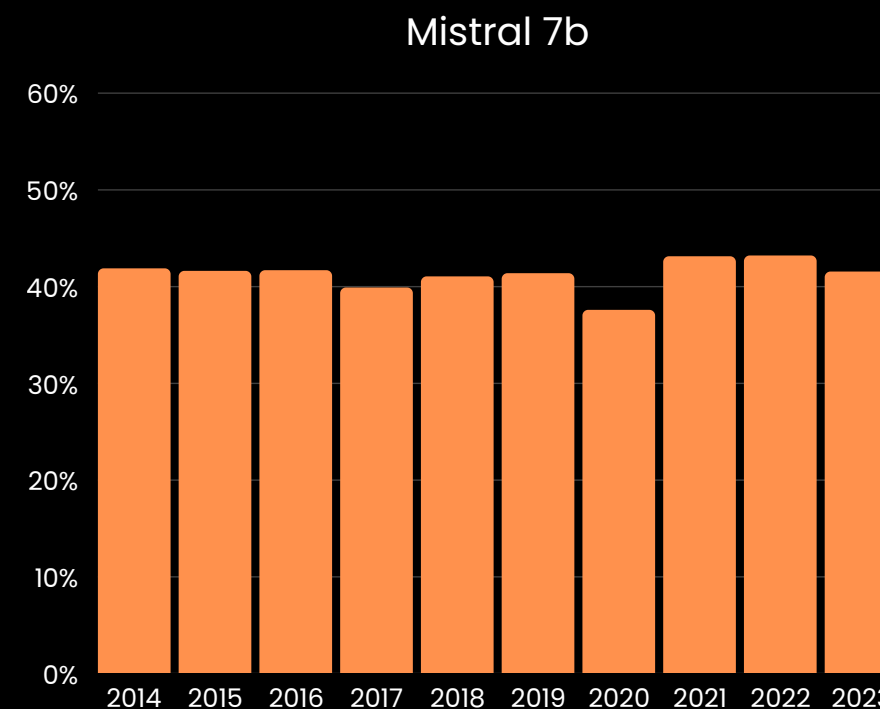
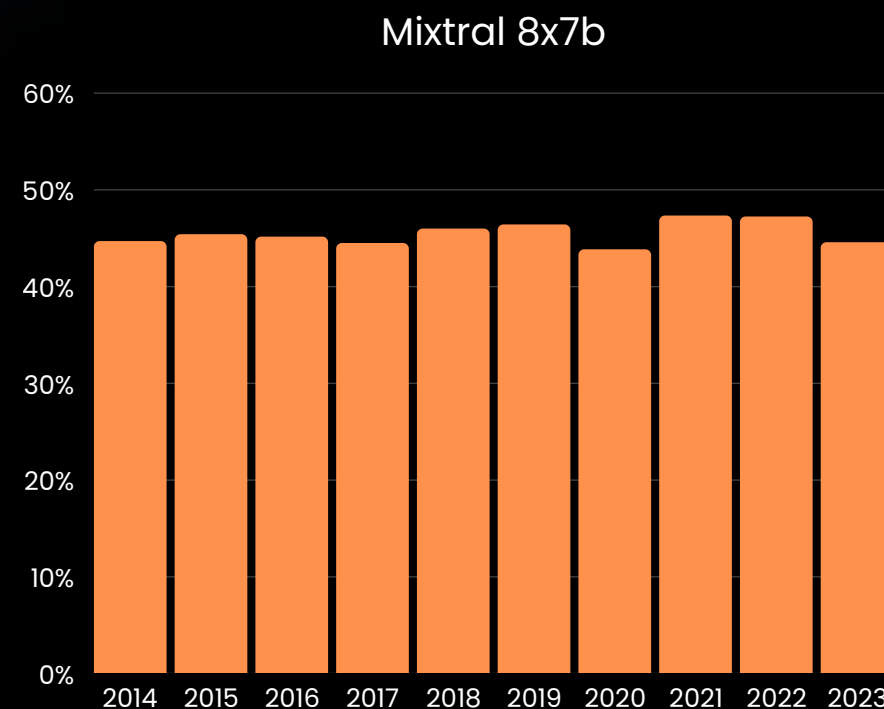
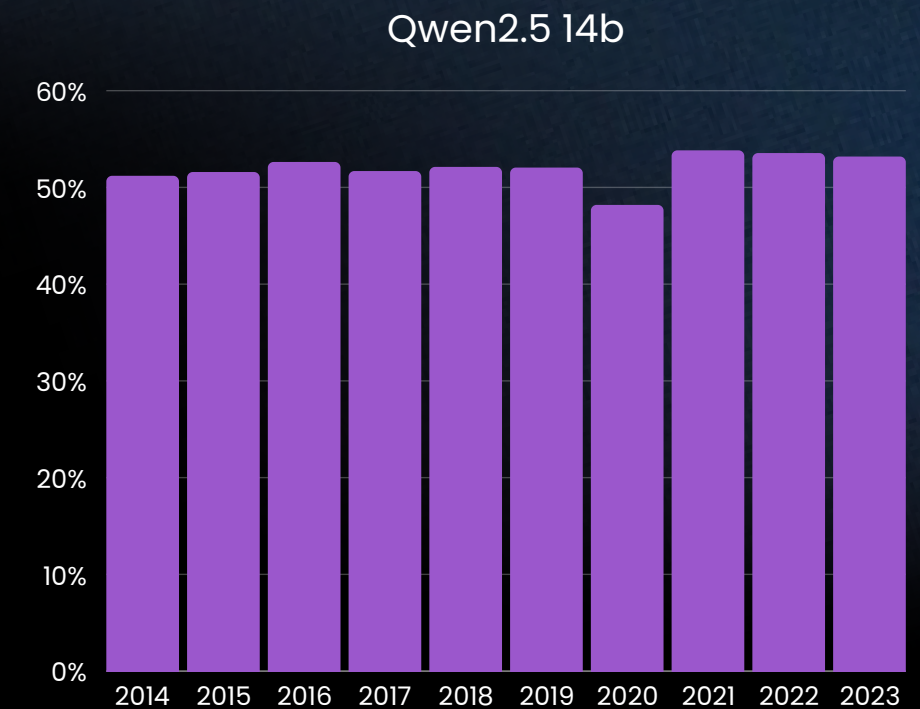
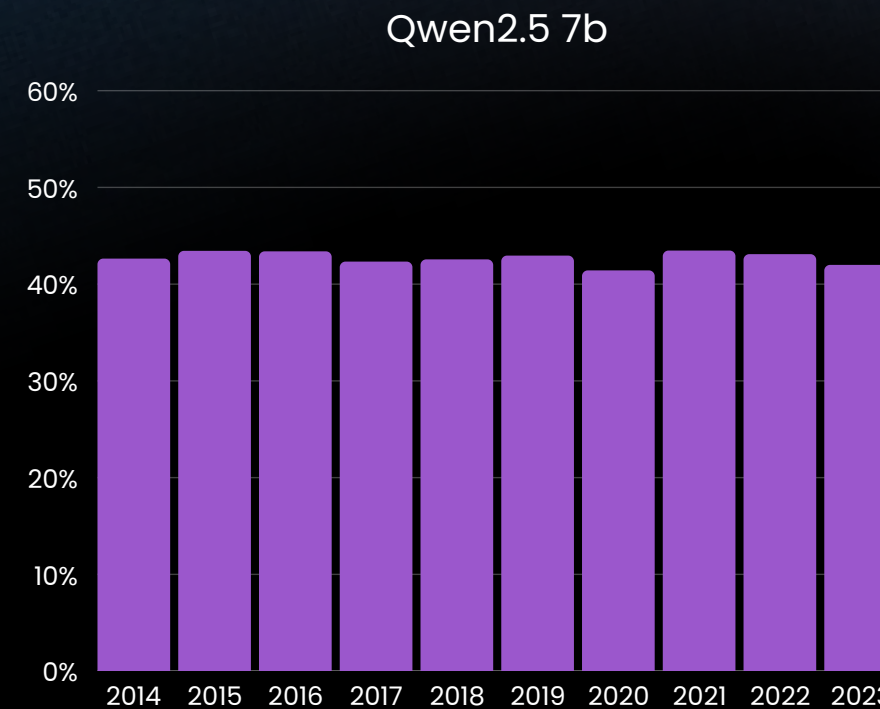
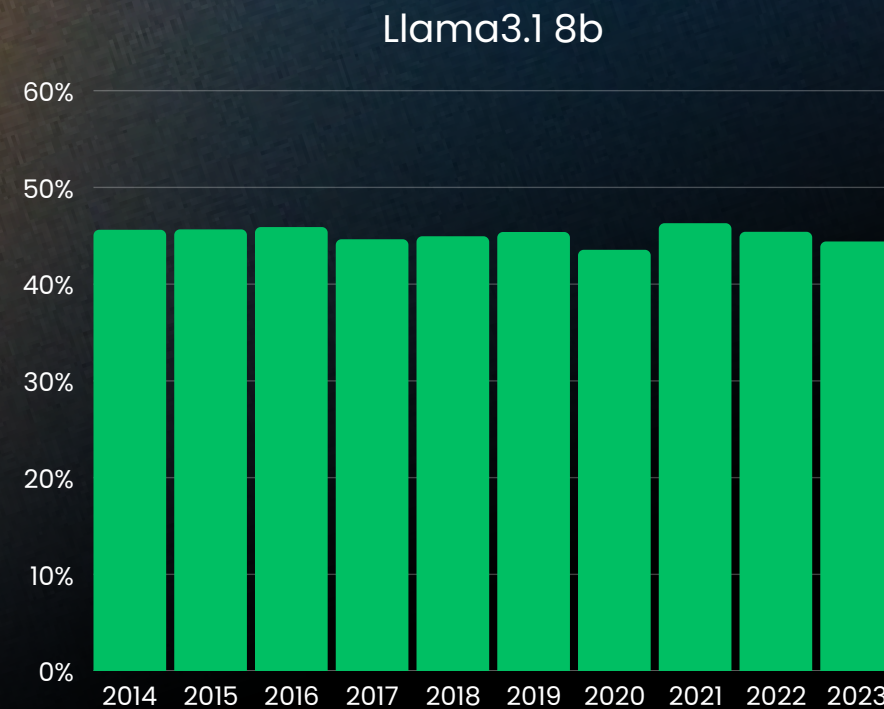
STRATEGIA TIME – PENULTIMATE

TOP 1 ACCURACY



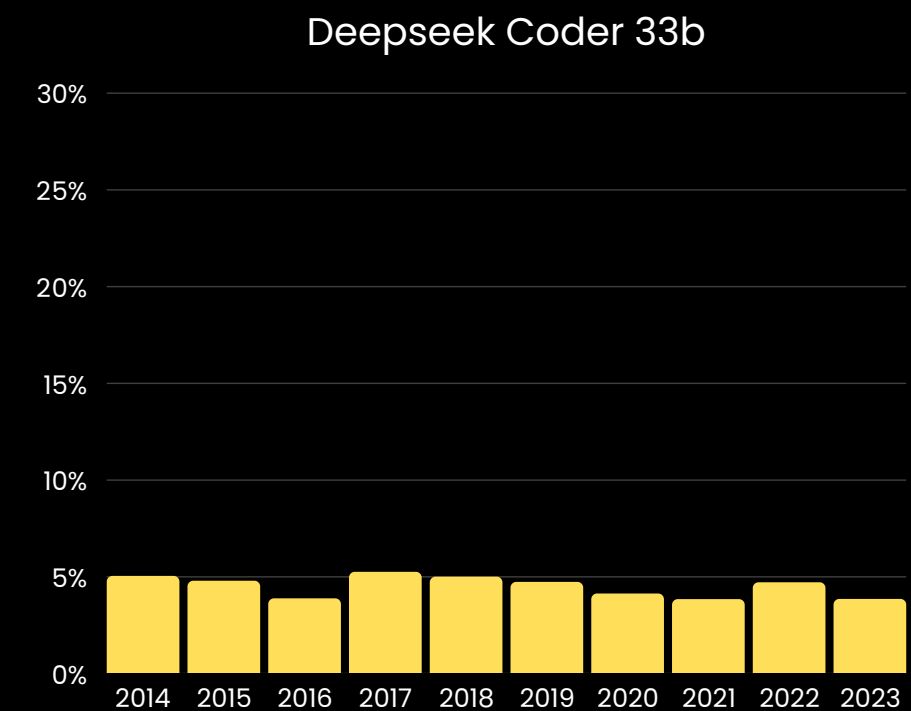
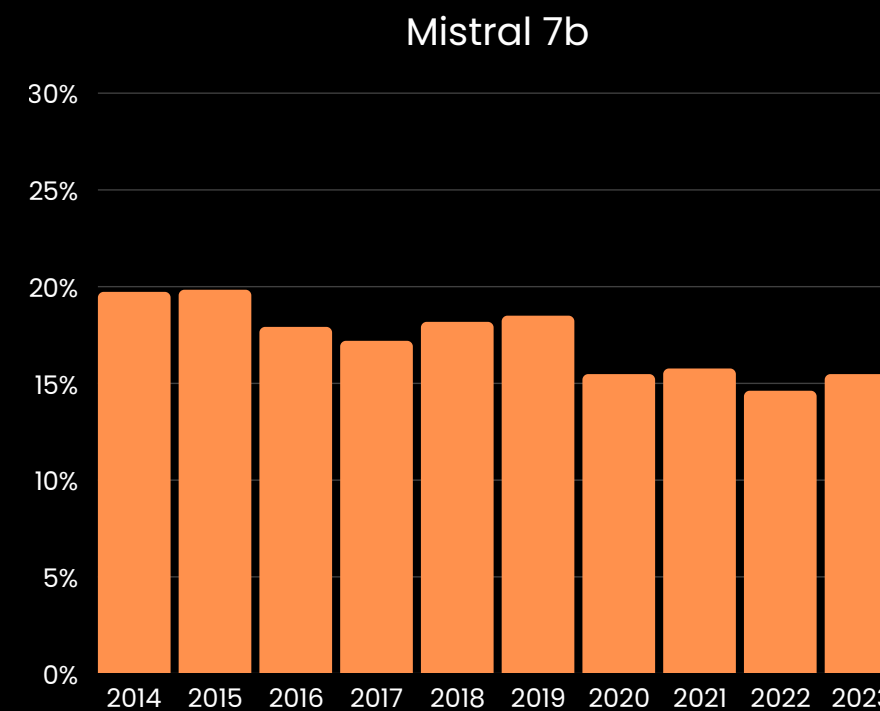
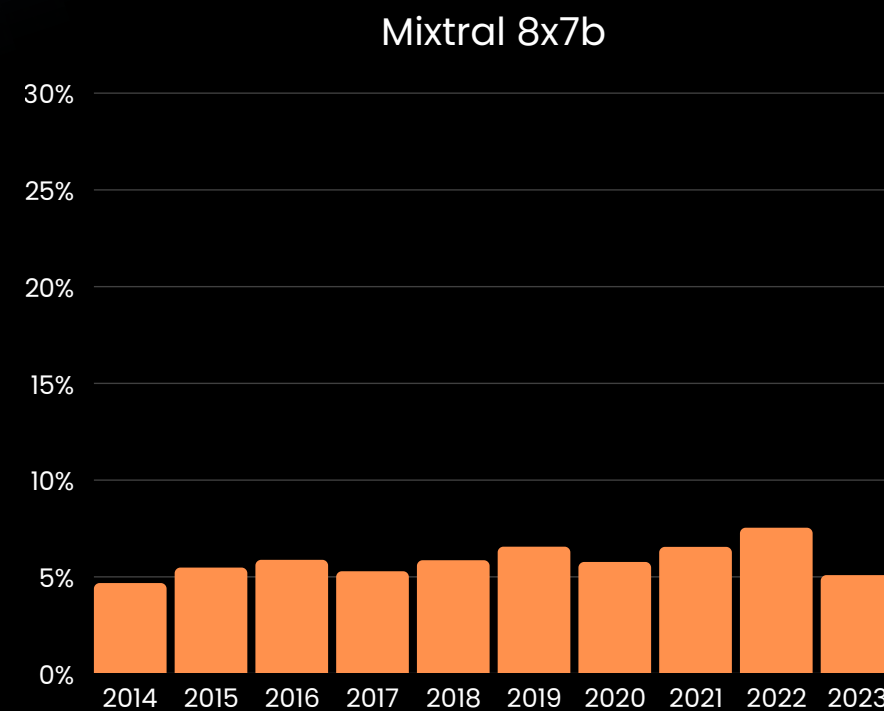
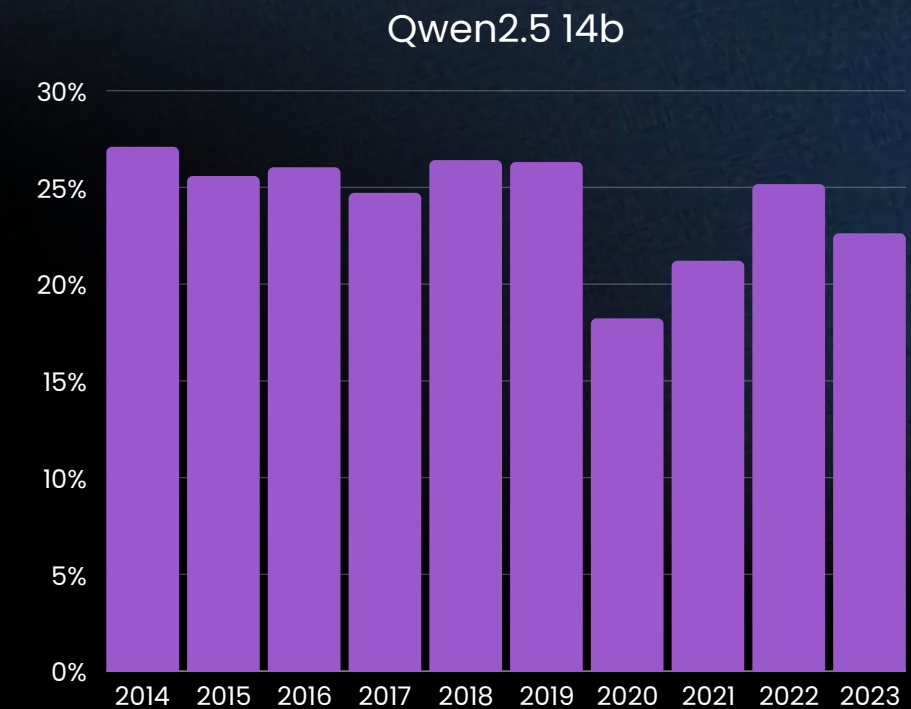
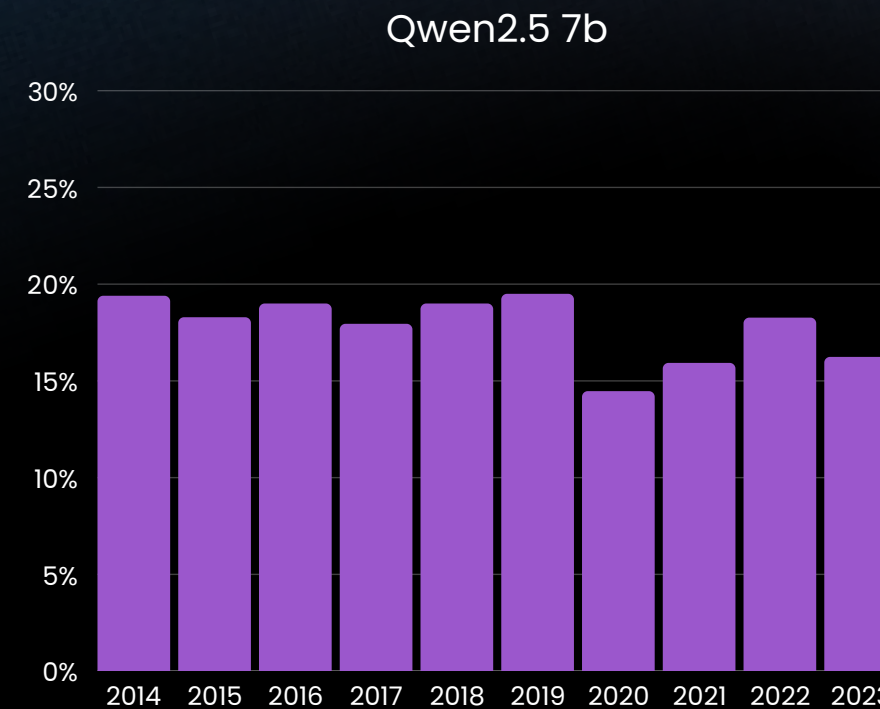
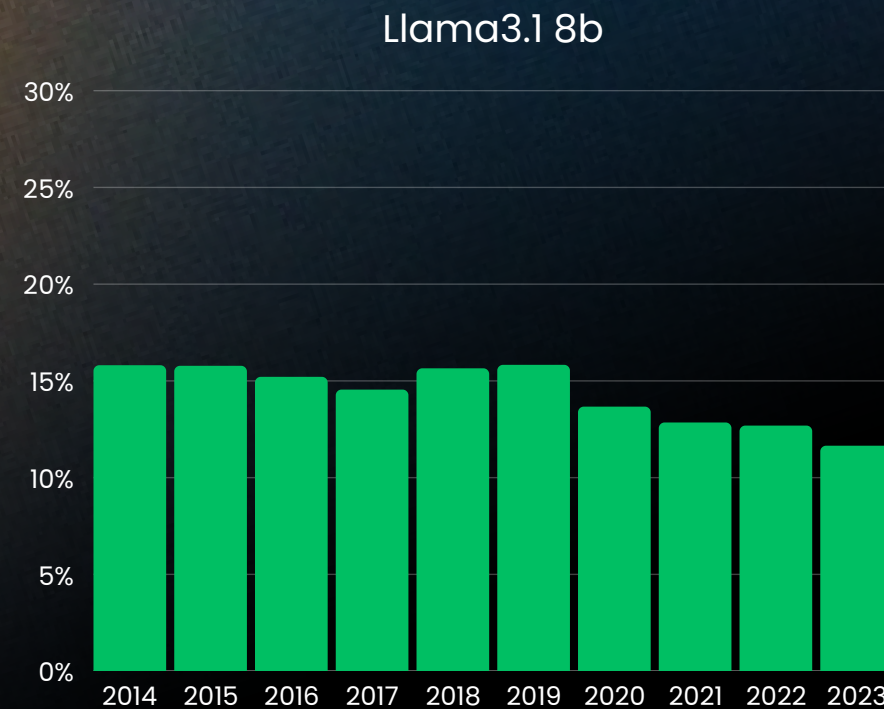
STRATEGIA TIME – PENULTIMATE

TOP-5 HIT RATE



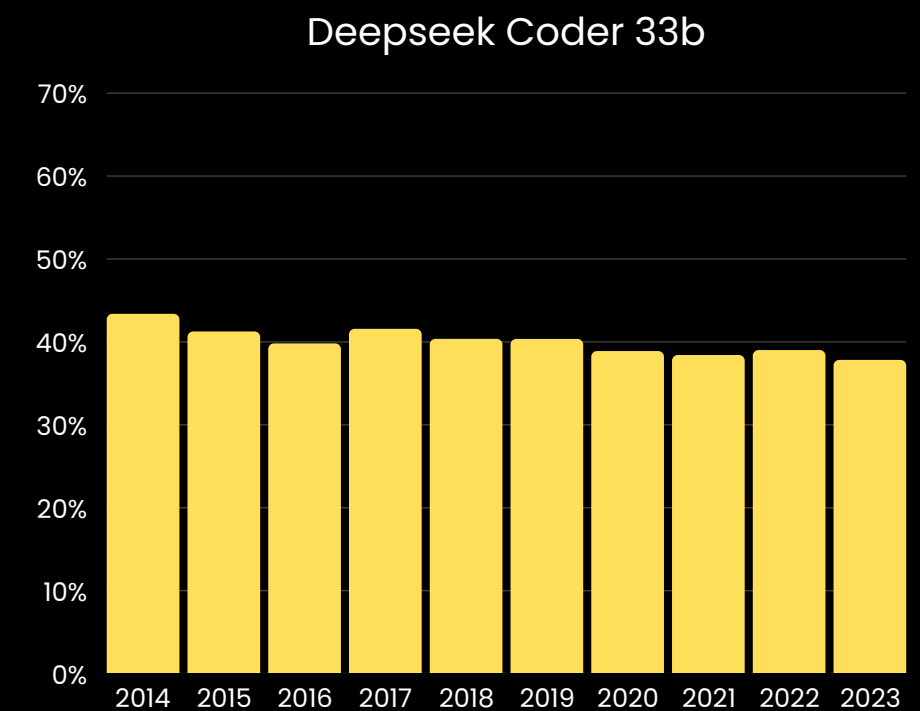
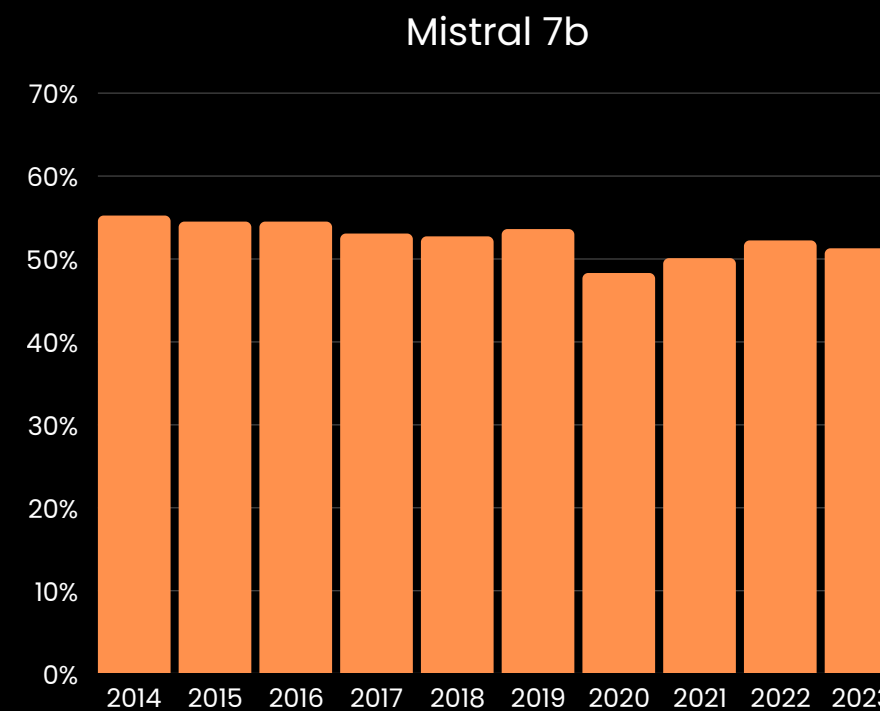
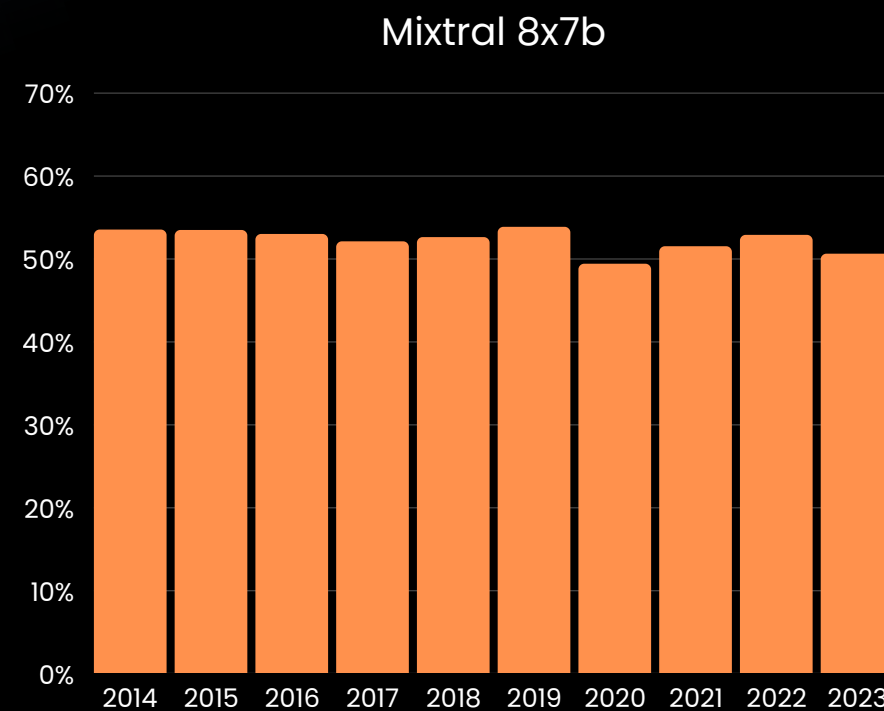
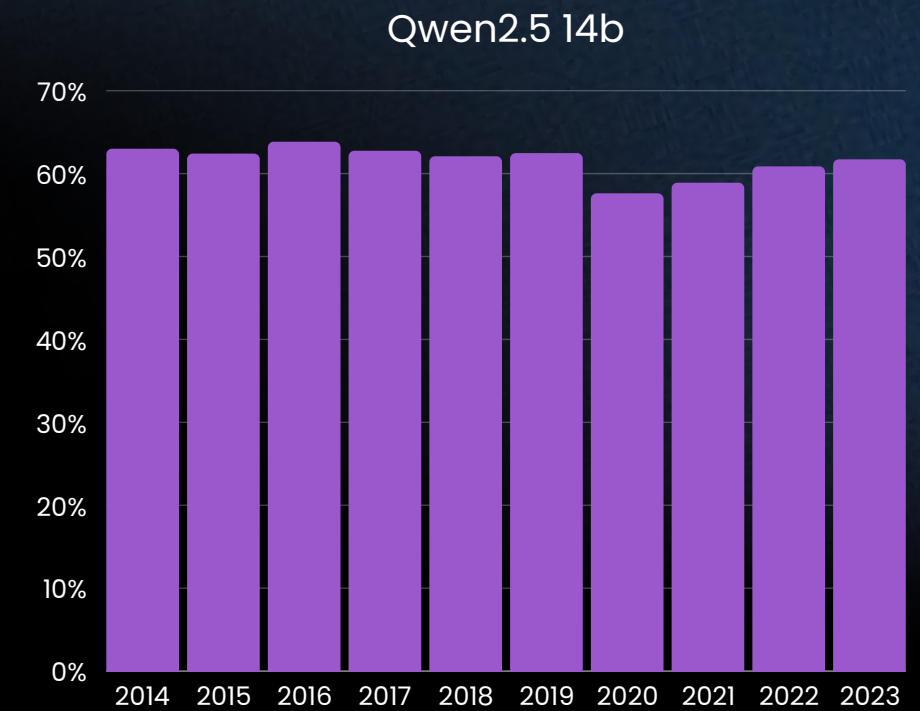
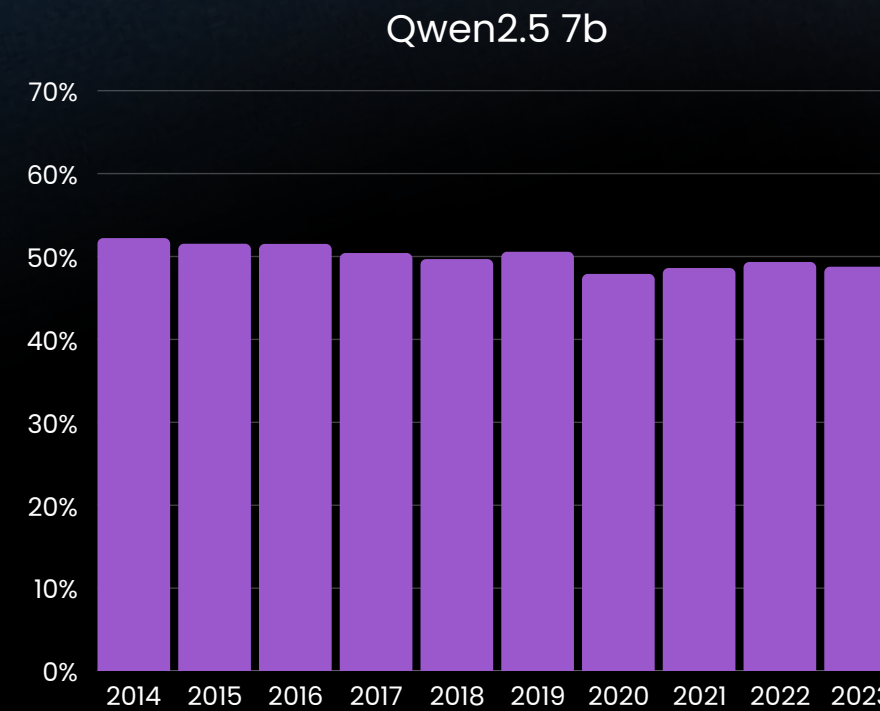
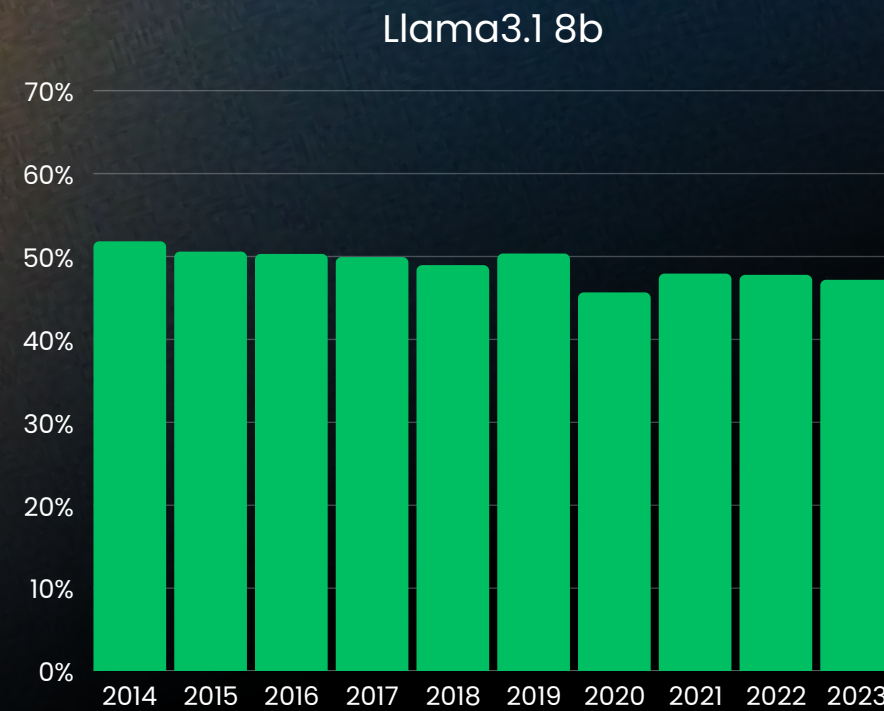
STRATEGIA TIME – MIDDLE

TOP 1 ACCURACY



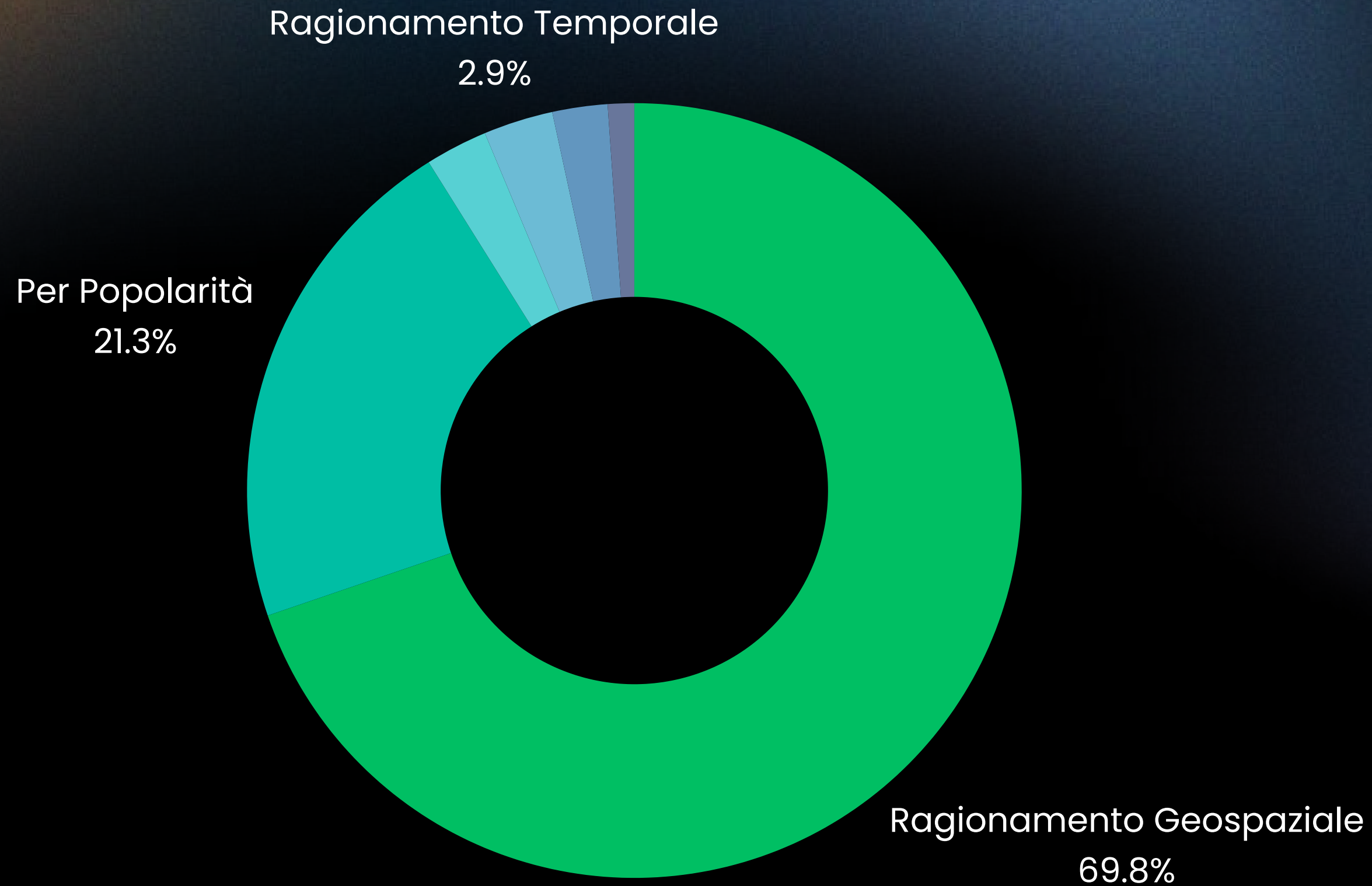
STRATEGIA TIME – MIDDLE

TOP-5 HIT RATE



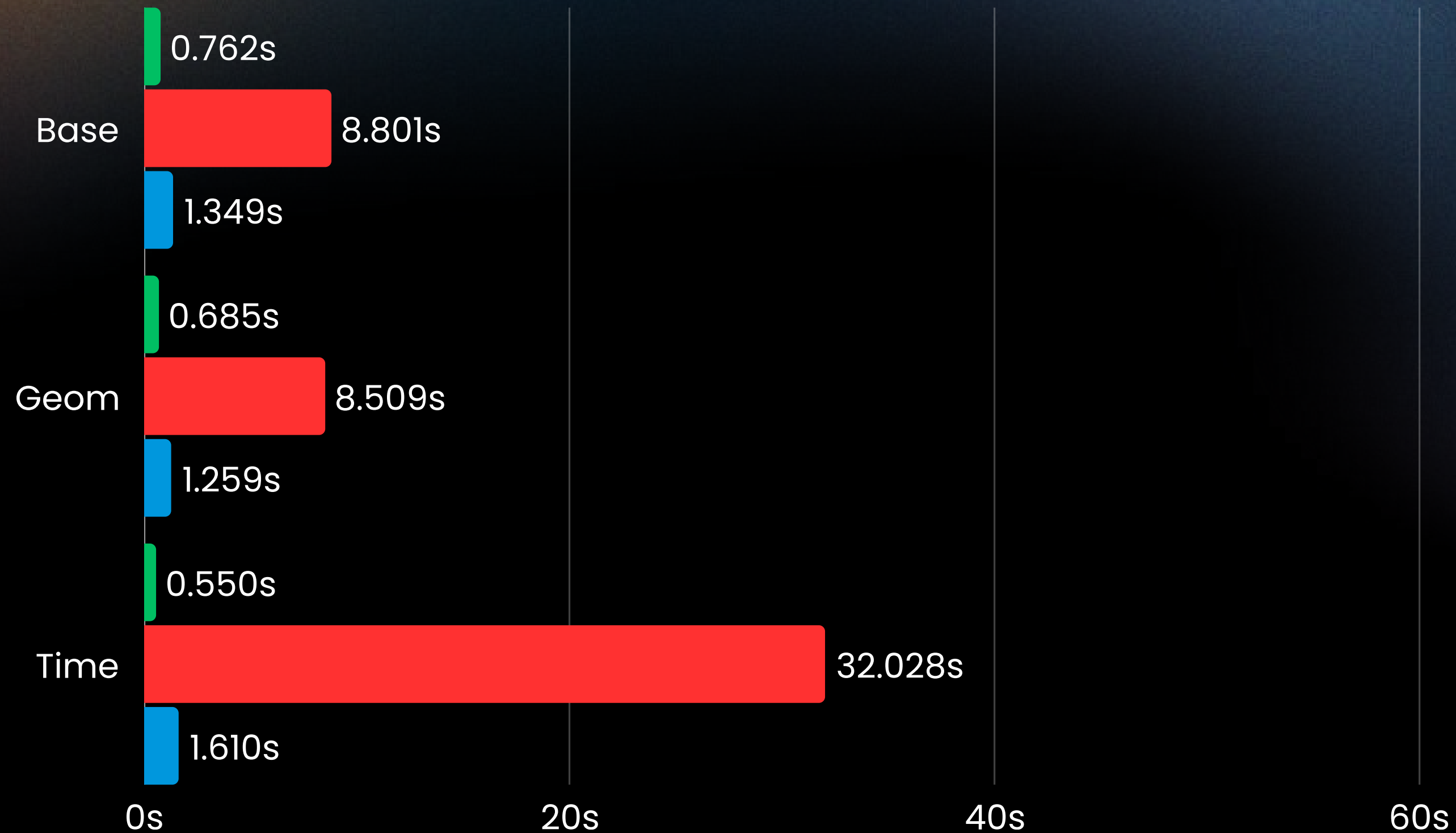
ANALISI CAMPO "MOTIVAZIONE"

Perchè l'LLM ha scelto quel POI?



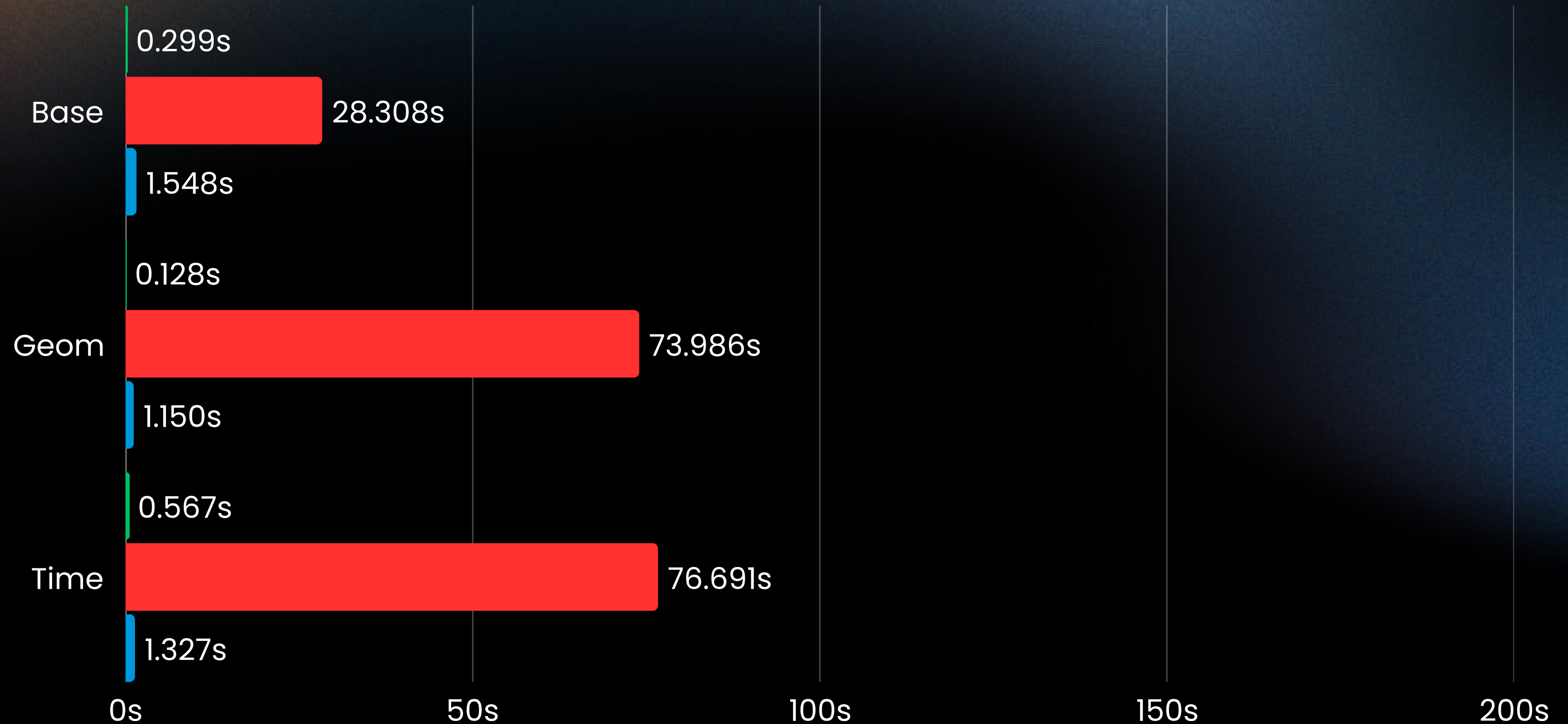
ANALISI DEI TEMPI DI RISPOSTA

LLAMA 3.1 8B



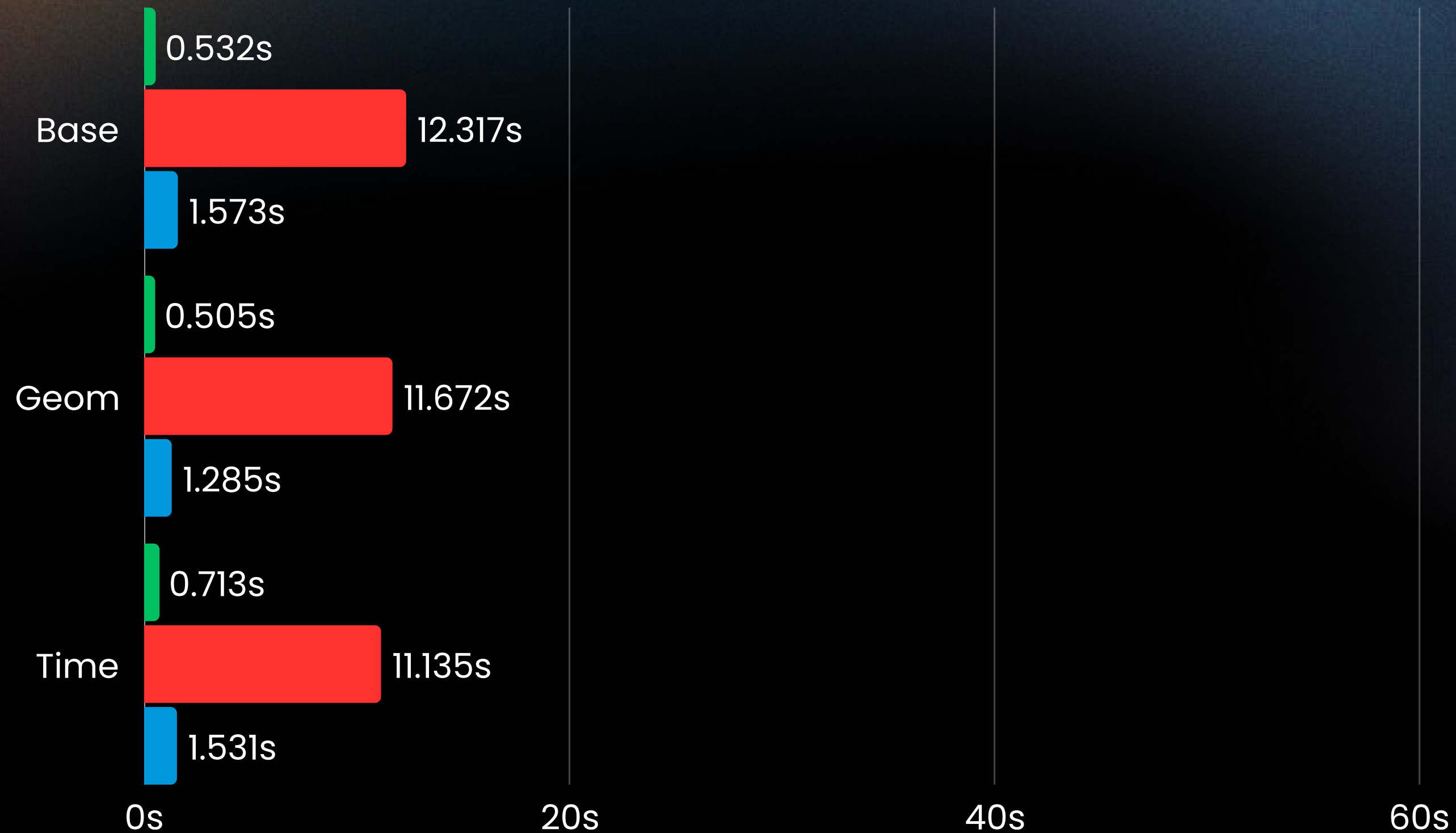
ANALISI DEI TEMPI DI RISPOSTA

QWEN 2.5 7B



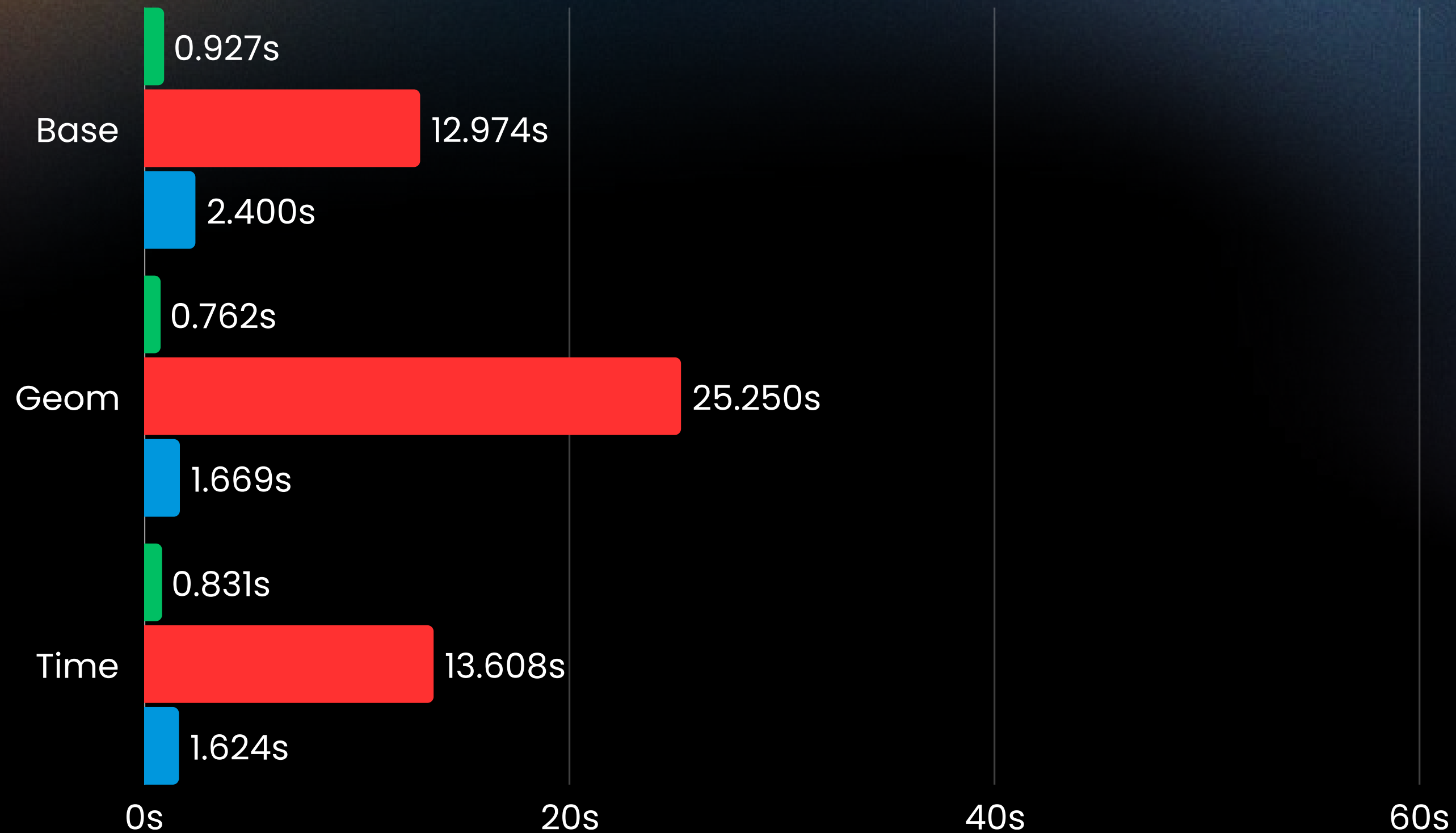
ANALISI DEI TEMPI DI RISPOSTA

MISTRAL 7B



ANALISI DEI TEMPI DI RISPOSTA

QWEN 2.5 14B



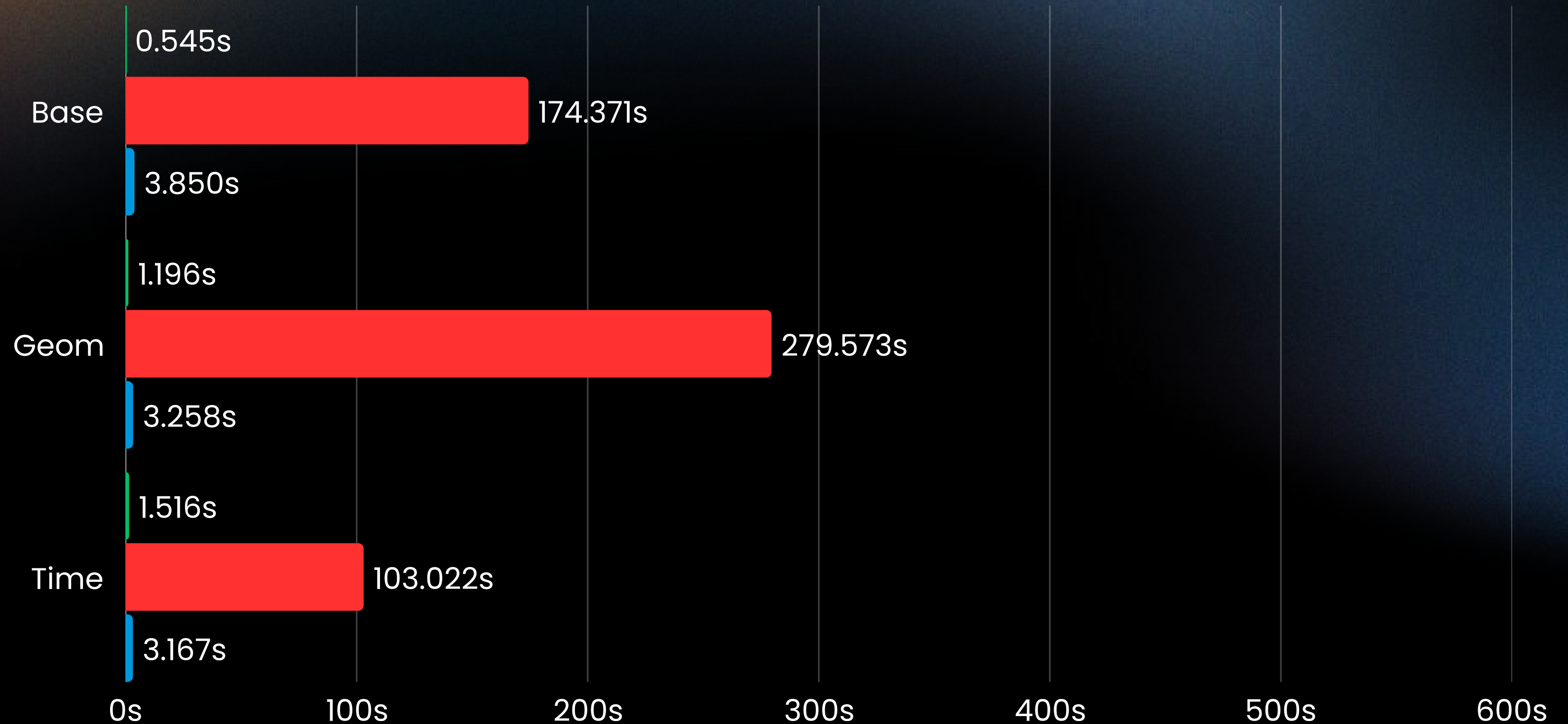
ANALISI DEI TEMPI DI RISPOSTA

MIXTRAL 8x7B

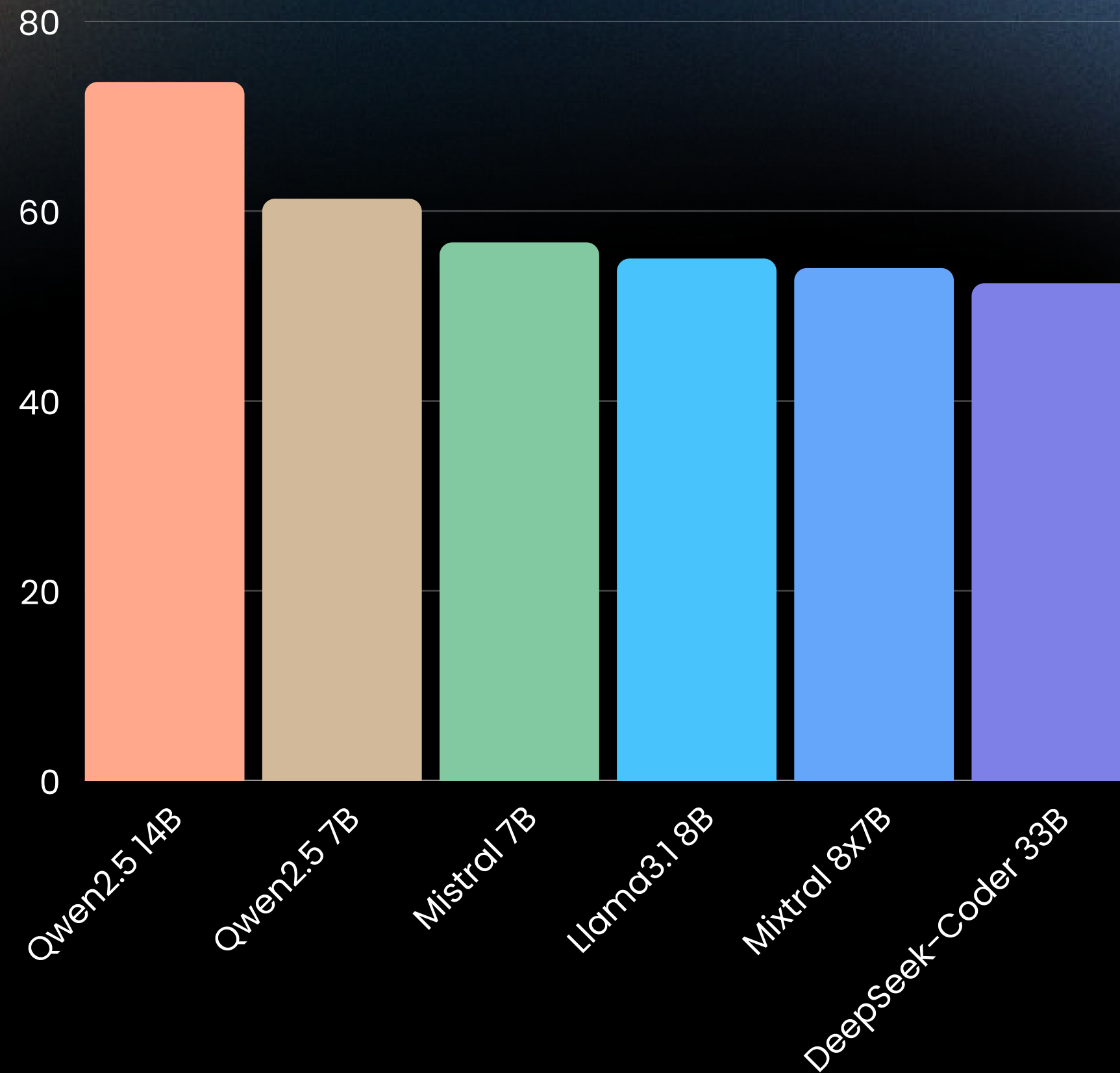


ANALISI DEI TEMPI DI RISPOSTA

DEEPSEEK CODER 33B



CONFRONTO INTER-MODELLO



CONCLUSIONE

08

IMPLEMENTAZIONI FUTURE



SISTEMA REAL TIME NELLE CITTÀ

Per distribuire uniformemente i turisti nei punti di interesse

GESTORE FLUSSI IMPIANTI SCIISTICI

Ottimizzazione distribuzione sciatori per evitare sovraffollamento

MIGLIORAMENTO DEL CONTESTO

Ad Esempio:

- Meteo
- Calendario
- Orari apertura/chiusura dei POI

SUPPORTO PER QUESTO IL LAVORO



CINECA

Accesso all'infrastruttura HPC Leonardo



UNIVERSITÀ DI VERONA

Accesso Al DataSet VeronaCard



PROF.SSA MIGLIORINI

Supervisione e guida durante lo sviluppo



AMICI

Sostegno nei momenti più critici



GENITORI

Supporto durante l'intero percorso universitario

"Se vuoi andare veloce, vai da solo. Se vuoi andare lontano, vai insieme"

GRAZIE PER L'ATTENZIONE

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Dettagli Implementativi

[Repository Git-Hub](#)